CY202 — Diagram Compendium

Every diagram in the Unit 1–5 question banks, with the question and the model answer it belongs to, plus a reading of the meeting summary

52 diagrams	**43** diagram-based questions	**5** units, no Unit 6	**145** questions parsed

What is in here

Part 1 — Findings from the meeting summary. What the briefing actually commits the professor to, and — checked line by line against the five banks — the five topics it flags that no question in your banks answers.

Part 2 — Figure index. All 52 diagrams listed with the question they illustrate and the bank page they come from.

Part 3 — Unit-wise compendium. For Units 1 to 5 in order: every diagram-based question, the diagram itself, and the model answer from the bank.

Part 4 — Draw-from-memory checklist. The diagrams most worth being able to reproduce by hand.

Every diagram is lifted from your own question banks at 190 dpi, so it prints cleanly at the size shown. Where a bank places two diagrams under one question, both are reproduced. Answers are the banks' own model answers, unabridged apart from removing page furniture.

Part 1 · Findings from the meeting summary

The file — *summary of the meeting in detail without losing a...* — is a three-page debrief of a class discussion with the professor, not a question paper. It is organised in four moves: the examination pattern, the numerical topics, a unit-wise priority list, and a closing set of two-mark topics. Read against your syllabus and banks, here is what it tells you and what it leaves open.

1. The paper pattern it commits to

- Theory and application questions against mathematical ones at **70:30 to 75:25**, so numericals are **20–25%** of the paper.
- **2 to 3 numericals sit in the short/two-mark section** and a few more in the long-answer section — that is a heavier numerical load in the short section than the banks imply, since only Units 4 and 5 carry worked numericals there.
- The stated target is **99–100 marks**, reachable by answering to the merit of the question; explicitly, no 5–10 page answers.

2. The four topics it singles out

Only four topics carry a verbal flag in the document, and those flags are the closest thing it contains to a prediction: the **nitrogen cycle** ("most important and tricky, 4–5 steps"), **characteristics of living beings** ("potential long question"), the **greenhouse effect and global warming** ("long question format"), and **Noise Standards 2000** ("expected as a 2-mark short question"). Each of the first three is well served by your banks — Unit 3 Q6, Unit 1 B1, Unit 4 Q3 and Q4.

3. Five flagged topics that no bank question answers

This is the part worth acting on today. Checking each topic the briefing names against the question text of all five banks, five of them have no matching question and, in most cases, no matching answer text either:

Flagged topic	What the banks actually contain
Coagulation, flocculation and sedimentation — their mechanisms	The words *coagulation* and *flocculation* do not appear anywhere in the Unit 5 bank; *sedimentation* appears once, inside the answer to B1. The briefing names all three as a mechanism question.
Disinfection and sterilisation techniques	Mentioned once, in the closing line of the B1 answer. There is no question and no worked answer on disinfection methods.
Noise Standards 2000 — definition and limit values	The Unit 4 bank never mentions the standards; it covers noise as a short note (Q8) and the sound-level formulas, but not the statutory limits by zone.
Metal centres — Mg in chlorophyll, Co in Vitamin B12, Ca^{2+} (+2) in Vitamin D3/carboxylase	Only magnesium appears, incidentally inside the chlorophyll answer (A26). Cobalt appears nowhere; B12 appears only as a question number in Unit 1.
Hydrocarbon discharge and vehicle-emission calculations	The briefing points to "the extra question set provided in the document from HOD". That set is not among the files here, and the Unit 4 bank carries only the two sound-level numericals.

Three further topics are present in the banks but thinner than the briefing describes them: **types of vaccines** (Unit 2 Q9 covers how vaccines work and their importance, not the typology), **7–8 types of pesticides** (Unit 2 Q6 asks only for four advantages and four disadvantages), and a **direct comparison of primary and tertiary water treatment** (Unit 5 B1 covers the three phases and the trickling filter, but never sets primary against tertiary).

4. What the document does not say

There is no unit-wise marks weightage, no blueprint of how many questions come from each unit, and no mention of a Unit 6 — consistent with your syllabus, which ends at Unit 5. So the

briefing tells you what to prioritise within a unit, but not how much of the paper any unit is worth; plan by topic density, which is why Unit 2's ten long answers and Unit 4's twenty short ones carry the most recall load.

5. What to do about it

The five gaps are the highest-risk items you hold: named by the professor, not drilled by the banks. Write those five answers yourself from the reference books in the syllabus — Fisher's *Environmental Biology* and Kiely's *Environmental Engineering* cover the water-treatment mechanisms and the noise standards respectively — rather than waiting for a bank question that does not exist.

Part 2 · Figure index

All diagrams in the five banks, in unit order. “Question” is the question whose answer the diagram sits under; “Bank page” is the page of the source PDF, so you can go back to it.

Fig	Unit	Q · marks	Diagram	Bank p.
1	U1	A17 · 2m	Fig. A17 — Phospholipid/fatty acid molecule and the resulting lipid bilayer / micelle	4
2	U1	A19 · 2m	Fig. A19 — Structures of the 20/21 proteinogenic amino acids grouped by side-chain type, with three-letter and…	4
3	U1	B2 · 7m	Fig. B2 — Carbohydrates: monosaccharides, disaccharides and polysaccharides, from simple to complex	7
4	U1	B5 · 7m	Fig. B5a — Rough ER vs. Smooth ER	8
5	U1	B5 · 7m	Fig. B5b — Structure and functions of the Golgi apparatus	8
6	U1	B6 · 7m	Fig. B6 — Comparative structure of plant and animal cells	9
7	U1	B8 · 7m	Fig. B8 — Generalised eukaryotic cell showing major organelles	10
8	U1	B9 · 7m	Fig. B9 — Enzyme mechanism: substrate binding, catalysis and product formation (example: sucrose → glucose + fructose)	11
9	U1	B10 · 7m	Fig. B10 — The four levels of protein structure, illustrated with haemoglobin	11
10	U1	B11 · 7m	Fig. B11 — Peptide bond formation between two amino acids (dehydration condensation, releasing H2O)	12
11	U1	B13 · 7m	Fig. B13a — Purine and pyrimidine bases; the nucleotide sugar-phosphate backbone	13
12	U1	B13 · 7m	Fig. B13b — The DNA double helix showing antiparallel strands, base pairing and replication	13
13	U2	Q1 · 7m	Fig: Reversible modes of enzyme inhibition — competitive, non-competitive, and uncompetitive.	5
14	U2	Q2 · 7m	Fig: Pathway of alcoholic fermentation showing glycolysis, ATP/NADH cycling, and ethanol formation.	6
15	U2	Q3 · 2m	Fig: Biomass conversion routes into biofuels, biochar, pyrolytic oil, heat, and syngas.	7
16	U2	Q4 · 7m	Fig: Chloroplast structure showing the light-dependent reactions (thylakoid) and the light-independent Calvin Cycle…	8
17	U2	Q4 · 7m	Fig: The Calvin Cycle — carbon fixation, reduction, and regeneration of ribulose.	9
18	U2	Q6 · 7m	Fig: Major types of pesticides used in agriculture.	10
19	U2	Q8 · 7m	Fig: Main types of insect natural enemies — predators, parasites, and pathogens.	12
20	U2	Q9 · 7m	Fig: How vaccines work — training the immune system without causing disease.	13
21	U3	Q11 · 2m	Fig: A simple food chain — producer to final consumer/decomposer.	2
22	U3	Q1 · 7m	Fig: Levels of biological organization — individual, population, community, ecosystem, biome and biosphere.	4
23	U3	Q1 · 7m	Fig: Interaction of biotic and abiotic factors in an ecosystem, showing productivity, nutrient cycling and the…	4
24	U3	Q3 · 7m	Fig: Trophic levels and food web — from primary producers to tertiary consumers.	6
25	U3	Q4 · 7m	Fig: Energy transfer and loss across trophic levels (energy pyramid).	6
26	U3	Q5 · 7m	Fig: The Carbon Cycle.	7
27	U3	Q6 · 7m	Fig: The Nitrogen Cycle — fixation, ammonification, nitrification, assimilation and denitrification.	8
28	U3	Q8 · 7m	Fig: The Phosphorus Cycle.	9

Fig	Unit	Q · marks	Diagram	Bank p.
29	U3	Q8 · 7m	Fig: The Sulphur Cycle.	9
30	U3	Q9 · 7m	Fig: The Oxygen Cycle.	10
31	U3	Q10 · 2m	Fig: A schematic tolerance curve for a single species population existing on a single environmental gradient.	10
32	U3	Q11 · 7m	Fig: The Hydrologic (Water) Cycle.	11
33	U4	Q1 · 7m	Fig. 1 — The six common (criteria) air pollutants designated by the EPA under the Clean Air Act / NAAQS.	5
34	U4	Q1 · 7m	Fig. 2 — Classification of air pollutants into particulate matter (coarse/fine/ultrafine PM) and gaseous pollutants.	6
35	U4	Q2 · 7m	Fig. 3 — Formation cycle of acid rain: SO_2 and NO_2 react with atmospheric moisture to form H_2SO_4/HNO_3, which fall as…	7
36	U4	Q6 · 7m	Fig. 4 — Working of an electrostatic precipitator: charged metal plates attract negatively charged particulates via…	8
37	U4	Q7 · 7m	Fig. 5 — Flue gas desulfurization: (left) wet scrubbing sorption spray tower; (right) limestone-based FGD plant…	9
38	U4	Q8 · 7m	Fig. 6 — Typical loudness (dB) of common sounds, from breathing (0 dB) to a rocket launch (140+ dB).	10
39	U4	Q9 · 7m	Fig. 7 — A cyclone separator: the swirling gas stream forces coarse particles to the outer wall and down into a…	11
40	U4	Q9 · 7m	Fig. 8 — A baghouse: dirty air is blown through fabric filter bags; particulates are trapped while clean air exits at…	11
41	U4	Q10 · 2m	Fig. 9 — Geosequestration: CO_2 captured from a power station is transported by pipeline and injected into underground…	12
42	U5	A5 · 2m	Fig. A5 - BOD exertion with time; BOD5 is read at day 5.	3
43	U5	A6 · 2m	Fig. A6 - Stages of eutrophication of a water body.	3
44	U5	A9 · 2m	Fig. A9 - Linear economy versus circular economy.	4
45	U5	B1 · 7m	Fig. B1a - The phases of wastewater treatment.	5
46	U5	B1 · 7m	Fig. B1b - Cross-section of a trickling filter.	6
47	U5	B1 · 7m	Fig. B1c - The activated sludge process (a second secondary-treatment method).	6
48	U5	B2 · 7m	Fig. B2 - Cross-section of a biological oxidation (stabilization) pond.	7
49	U5	B3 · 7m	Fig. B3 - The four functional elements of integrated solid waste management.	8
50	U5	B4 · 7m	Fig. B4 - Major components of a hazardous waste management plan.	9
51	U5	B5 · 7m	Fig. B5 - Steps in e-waste recycling.	10
52	U5	B7 · 7m	Fig. B7 - EM radiation field regions around an antenna.	11

Part 3 · Unit-wise compendium of diagrams and diagram-based questions

Each entry gives the question as it appears in the bank, its marks, the diagram, and the bank's own model answer. Questions without a diagram are not repeated here — this part is only the diagram-based ones.

UNIT 1 · Cells and Biomolecules — Structure

10 diagram-based questions · 12 diagrams in this unit

QA17. Draw the bilayer structure of a lipid

2 marks · from the Unit 1 bank

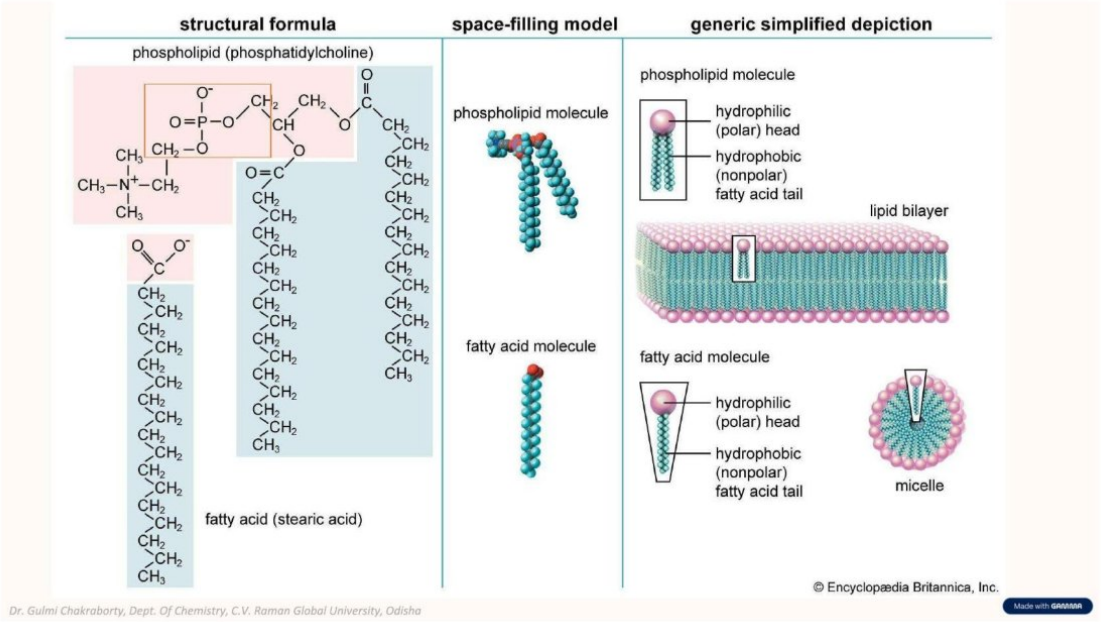


Fig. 1 — Fig. A17 — Phospholipid/fatty acid molecule and the resulting lipid bilayer / micelle *(bank p. 4)*

Answer. A phospholipid bilayer is formed of two layers of phospholipid molecules arranged tail-to-tail. The hydrophilic (polar) phosphate heads face outward toward the aqueous medium on both sides, while the hydrophobic (non-polar) fatty-acid tails face inward, shielded from water. This amphipathic self-assembly forms the structural basis of every biological membrane.

QA19. Give an example each of an acidic, basic, neutral, sulphur-containing, and aromatic amino

2 marks · from the Unit 1 bank

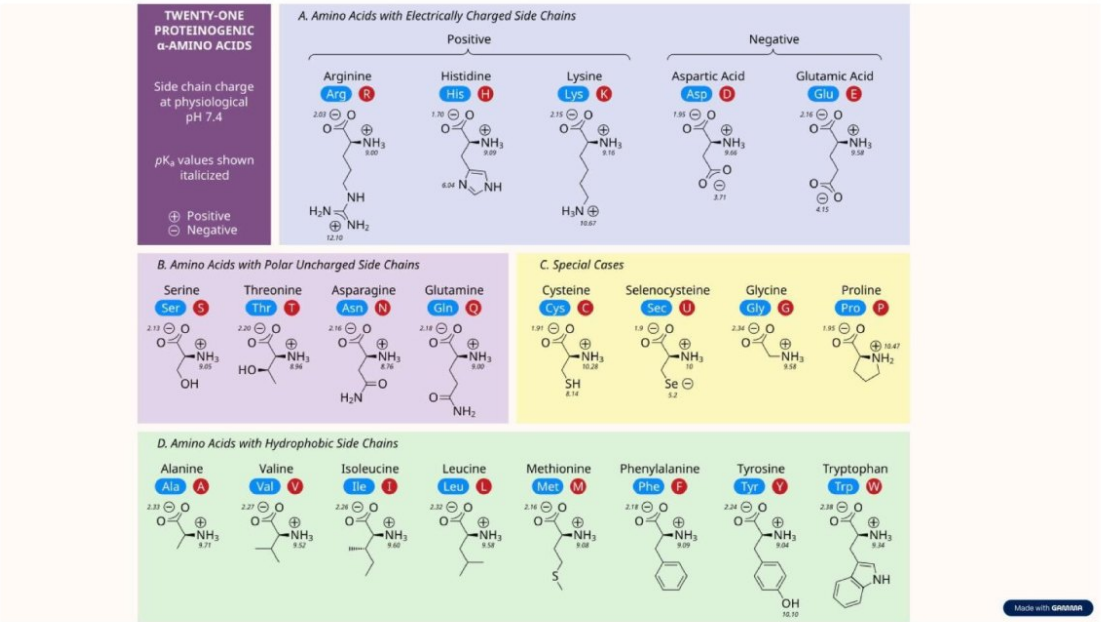


Fig. 2 — Fig. A19 — Structures of the 20/21 proteinogenic amino acids grouped by side-chain type, with three-letter and single-letter codes (bank p. 4)

Answer. acid. Mention their three-letter and single-letter codes. Ans: Acidic: Glutamic acid (Glu, E). Basic: Lysine (Lys, K). Neutral/polar-uncharged: Serine (Ser, S). Sulphur-containing: Cysteine (Cys, C). Aromatic: Phenylalanine (Phe, F). Structures are shown below.

QB2. Elaborate on the different types of organic constituents present in living cells, with examples

7 marks · from the Unit 1 bank

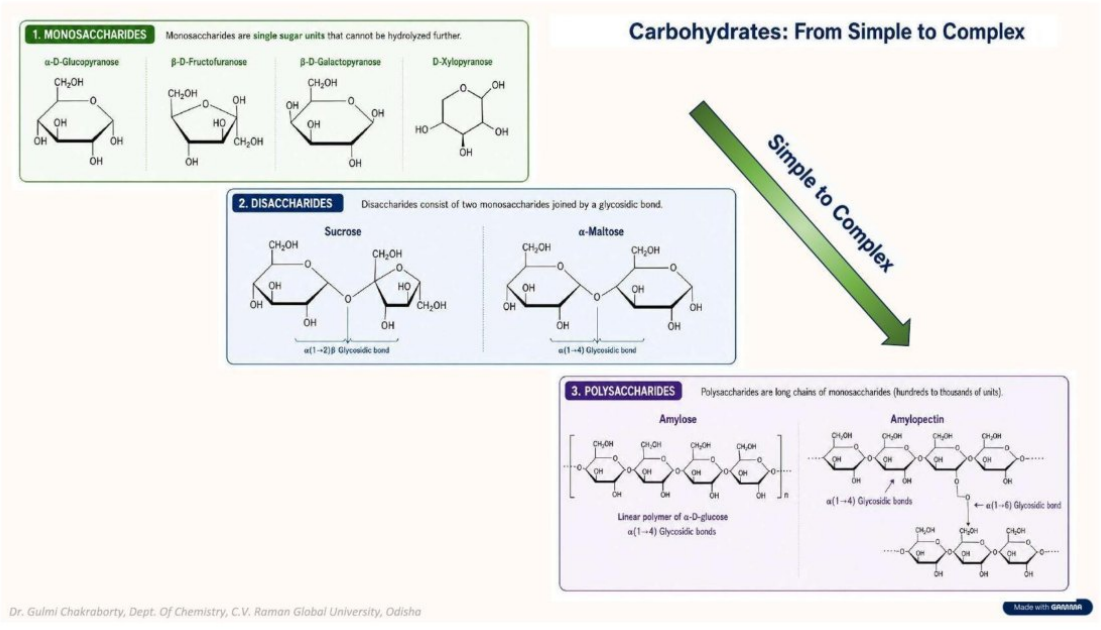


Fig. 3 — Fig. B2 — Carbohydrates: monosaccharides, disaccharides and polysaccharides, from simple to complex (bank p. 7)

Answer. Living cells are composed chiefly of three classes of organic macromolecules — carbohydrates, lipids and proteins — together with nucleic acids, each serving distinct structural and functional roles. Carbohydrates: polyhydroxy aldehydes/ketones with general formula Cn(H2O)n. Classified by degree • of polymerisation into monosaccharides (glucose, fructose, galactose), disaccharides (sucrose, lactose, maltose), oligosaccharides (2–9 units, involved in cell recognition) and polysaccharides (starch, glycogen — energy storage; cellulose, chitin — structural support). Lipids: a structurally diverse group defined by solubility in non-polar solvents. Includes fatty acids, • triacylglycerols (energy storage), phospholipids (membrane structure), steroids (e.g., cholesterol — hormone precursor), and lipoproteins (lipid transport in blood). Proteins: polymers of 20 standard amino acids linked by peptide bonds; serve as enzymes, structural • scaffolds, transporters, receptors, antibodies and hormones. Structure is hierarchical — primary, secondary, tertiary and quaternary. Nucleic acids: DNA and RNA, polymers of nucleotides (base + pentose sugar + phosphate); store and • express genetic information.

QB5. Describe the structure and functions of cell organelles such as the endoplasmic reticulum,

7 marks · from the Unit 1 bank

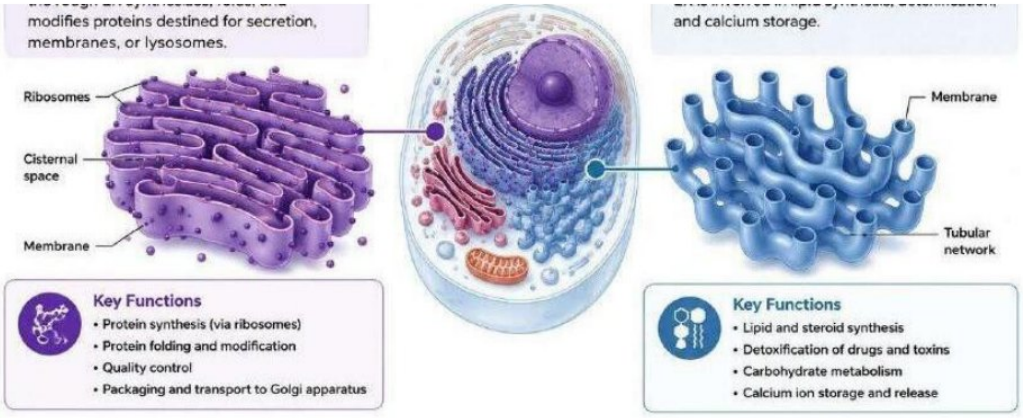


Figure 2.3: Structure of the Endoplasmic reticulum

The endoplasmic reticulum occurs in two structurally and functionally distinct forms. The Rough Endoplasmic

Fig. 4 — Fig. B5a — Rough ER vs. Smooth ER (bank p. 8)

membrane. With the exception of sperm cells and red blood cells, the endoplasmic reticulum is present in essentially every type of eukaryotic cell.

2.6 The Golgi Apparatus

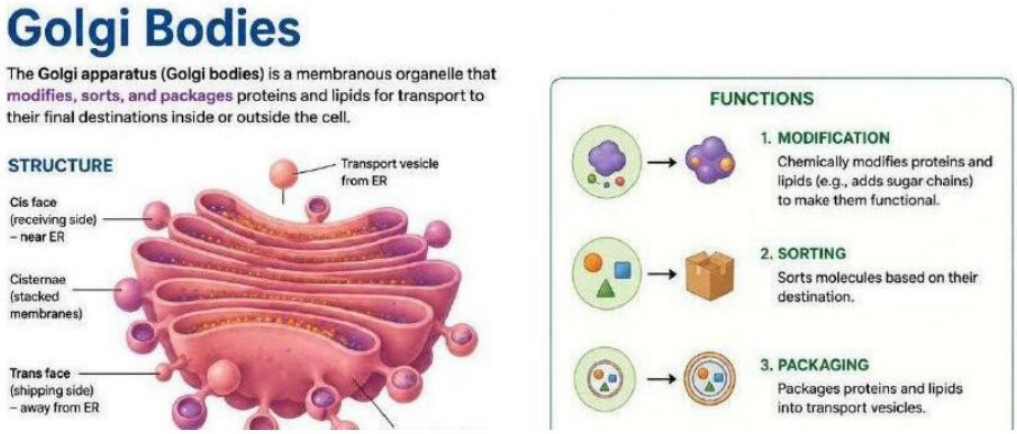


Fig. 5 — Fig. B5b — Structure and functions of the Golgi apparatus (bank p. 8)

Answer. Golgi apparatus, mitochondria and lysosomes. Endoplasmic Reticulum (ER): A network of membrane-bound sacs and tubules occurring in two forms. Rough ER (RER) is studded • with ribosomes and is the principal site of protein synthesis, folding and quality control. Smooth ER (SER) lacks ribosomes and carries out lipid/steroid synthesis, detoxification of drugs/toxins, and calcium storage. Golgi Apparatus: A stack of membranous cisternae (cis face receives from ER, trans face ships out) that modifies (adds • sugar chains), sorts, packages, and delivers proteins/lipids to the plasma membrane, lysosomes, or for secretion. Mitochondria: Double-membraned organelles that are the principal site of ATP generation via aerobic respiration — • the "powerhouse of the cell"; contain their own DNA and ribosomes. Lysosomes: Membrane-bound vesicles containing hydrolytic enzymes for intracellular digestion of waste and • foreign material; termed "suicide bags" because rupture of the membrane can digest the cell itself (autolysis).

QB6. What are the differences between plant cells and animal cells?

7 marks · from the Unit 1 bank

A recurring examination and conceptual theme in cell biology is the systematic comparison of plant, animal, and bacterial cells. The following consolidated tables present these comparisons across structural, biochemical, and physiological criteria.

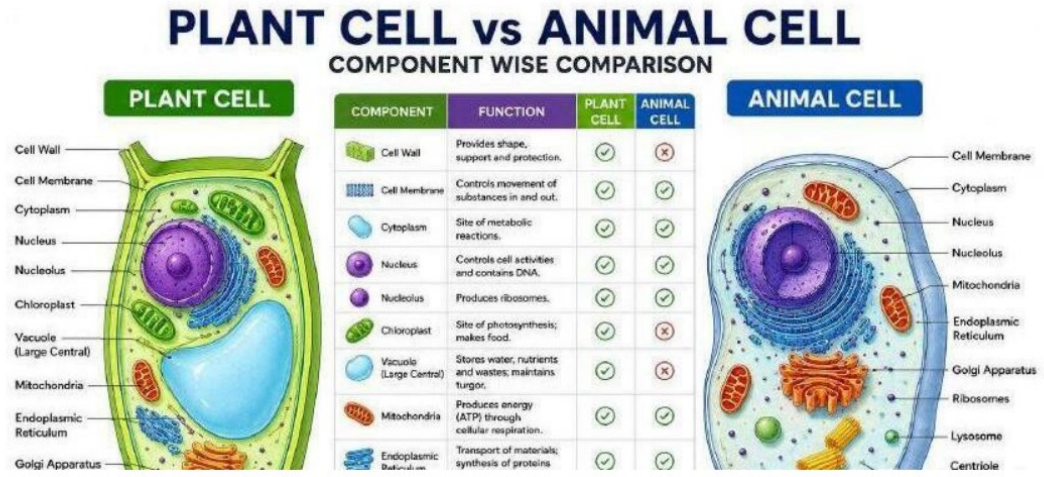


Fig. 6 — Fig. B6 — Comparative structure of plant and animal cells (bank p. 9)

Answer. Characteristic Plant Cell Animal Cell Cell wall Present (cellulose) Absent Shape Fixed, rectangular Flexible, round/oval Chloroplast Present Absent Vacuole Large, single, central Small or absent Centriole Absent Present Lysosomes Few / rare Present Nutriti

QB8. Define cell. Write down five characteristics of cells. Draw a structure of the cell with a labelled

7 marks · from the Unit 1 bank

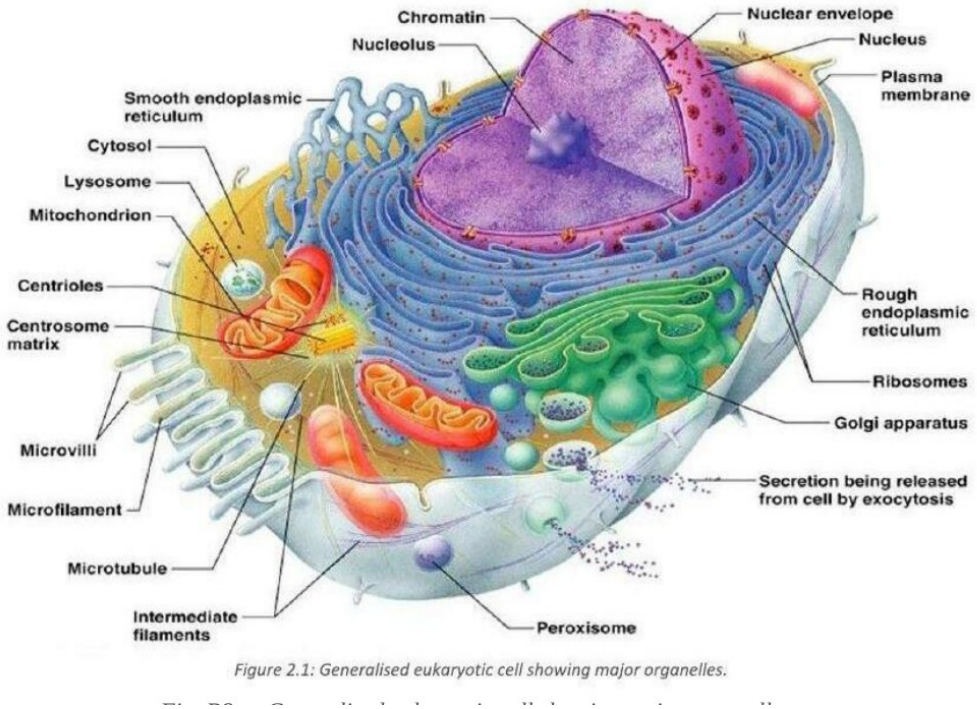


Figure 2.1: Generalised eukaryotic cell showing major organelles.

Fig. 7 — Fig. B8 — Generalised eukaryotic cell showing major organelles (bank p. 10)

Answer. diagram. Ans: A cell is the smallest structural and functional unit of life, capable of independent existence and carrying out all essential life processes. Five characteristics of cells: Cells are the lowest level of structural organisation common to every living system, and house the • organism's hereditary material (DNA). Cells range enormously in size — from ~0.1 µm (Mycoplasma) to ~150 mm (ostrich egg) — yet share a • common organisational plan. Cells may be prokaryotic (no true nucleus) or eukaryotic (membrane-bound nucleus). • Cells carry out metabolism, respond to stimuli, grow, and reproduce. • Cells contain specialised organelles, each performing a distinct function essential to the cell's survival.

QB9. What are enzymes? Discuss their structure, characteristics and working mechanism in the

7 marks · from the Unit 1 bank

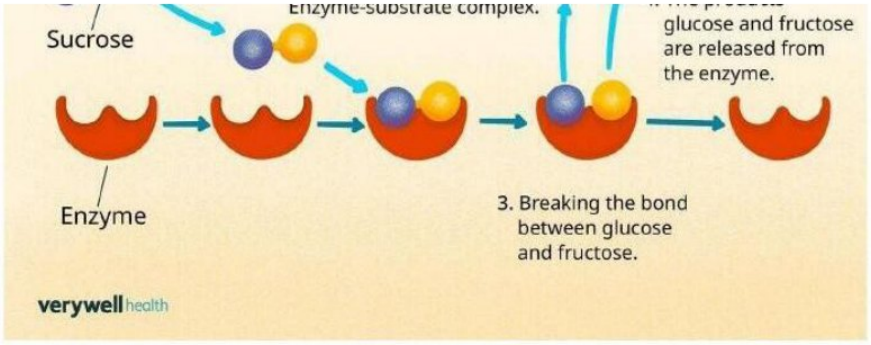


Figure 5.2: Enzyme mechanism — substrate binding, catalysis, and product formation.

Step 1 — Substrate Binding: At the active site, the enzyme binds to a specific substrate molecule, forming an enzyme–substrate complex.

Step 2 — Catalysis: The enzyme facilitates the chemical transformation, either breaking down the substrate (catabolism) or synthesising a new product (anabolism).

Fig. 8 — Fig. B9 — Enzyme mechanism: substrate binding, catalysis and product formation (example: sucrose → glucose + fructose) *(bank p. 11)*

Answer. body. Ans: Enzymes are biocatalysts, predominantly proteinaceous, that accelerate biochemical reactions within living systems by lowering the activation energy, without being consumed in the process. Structure: A complete functional enzyme (holoenzyme) comprises an apoenzyme (the protein component, • catalytically inactive alone) and, in many cases, a cofactor (a non-protein component — a metal ion or organic molecule). A specific region called the active site is where the substrate binds and catalysis occurs. • Characteristics: Highly specific for their substrate (Lock-and-Key / Induced Fit model); accelerate reactions without • being consumed; sensitive to temperature and pH (e.g., optimal ~37°C for human enzymes; pepsin works best at pH 2); may require cofactors/coenzymes (e.g., Zn, Mg, NAD+); susceptible to competitive and non-competitive inhibition. Mechanism of action: Step 1 — Substrate binding: the enzyme binds a specific substrate at the active site, forming an • enzyme–substrate complex. Step 2 — Catalysis: the enzyme lowers the activation energy and facilitates the chemical transformation • of the substrate. Step 3 — Product formation: product(s) are released and the enzyme is regenerated unchanged, ready • for further reaction cycles. Physiological examples: Amylase (starch → maltose), Pepsin (proteins → peptides), Lipase (fats → glycerol + fatty acids).

QB10. Distinguish between primary, secondary, tertiary and quaternary structures of protein

7 marks · from the Unit 1 bank

Level	Description
Primary	The linear sequence of amino acids joined by covalent peptide bonds, read N-terminus to C-terminus; encoded directly by the gene.
Secondary	Local, regular folding patterns stabilised by hydrogen bonds between backbone -NH and -C=O groups: the α-helix (right-handed coil) and β-sheet (extended, laterally H-bonded strands).
Tertiary	The overall 3-D folding of the entire polypeptide chain, stabilised by hydrophobic interactions, hydrogen bonds, ionic bonds, disulphide bridges (-S-S-) and van der Waals forces; creates the active site.
Quaternary	The association of two or more polypeptide subunits into a single functional complex, e.g., haemoglobin (2α + 2β chains), stabilised by the same non-covalent forces plus inter-subunit disulphide bonds.

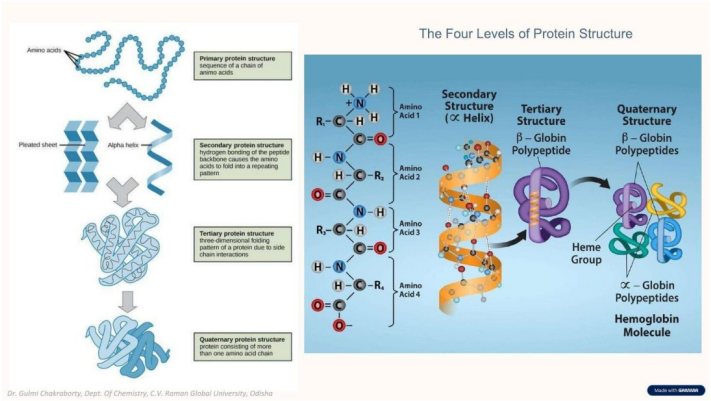


Fig. 9 — Fig. B10 — The four levels of protein structure, illustrated with haemoglobin (bank p. 11)

Answer. Protein structure is hierarchical — each level builds upon the previous, and disruption of any level (denaturation) typically abolishes biological function. Level Description Primary The linear sequence of amino acids joined by covalent peptide bonds, read N- terminus to C-terminus; encoded directly by the gene. Secondary Local, regular folding patterns stabilised by hydrogen bonds between backbone -NH and -C=O groups: the α-helix (right-handed coil) and β-sheet (extended, laterally H- bonded strands). Tertiary The overall 3-D folding of the entire polypeptide chain, stabilised by hydrophobic interactions, hydrogen bonds, ionic bonds, disulphide bridges (-S-S-) and van der Waals forces; creates the active site. Quaternary The association of two or more polypeptide subunits into a single functional complex, e.g., haemoglobin (2α + 2β chains), stabilised by the same non-covalent forces plus inter-subunit disulphide bonds.

QB11. Depict the primary structure of the following peptide codes: FYP, ATF, DCP

7 marks · from the Unit 1 bank

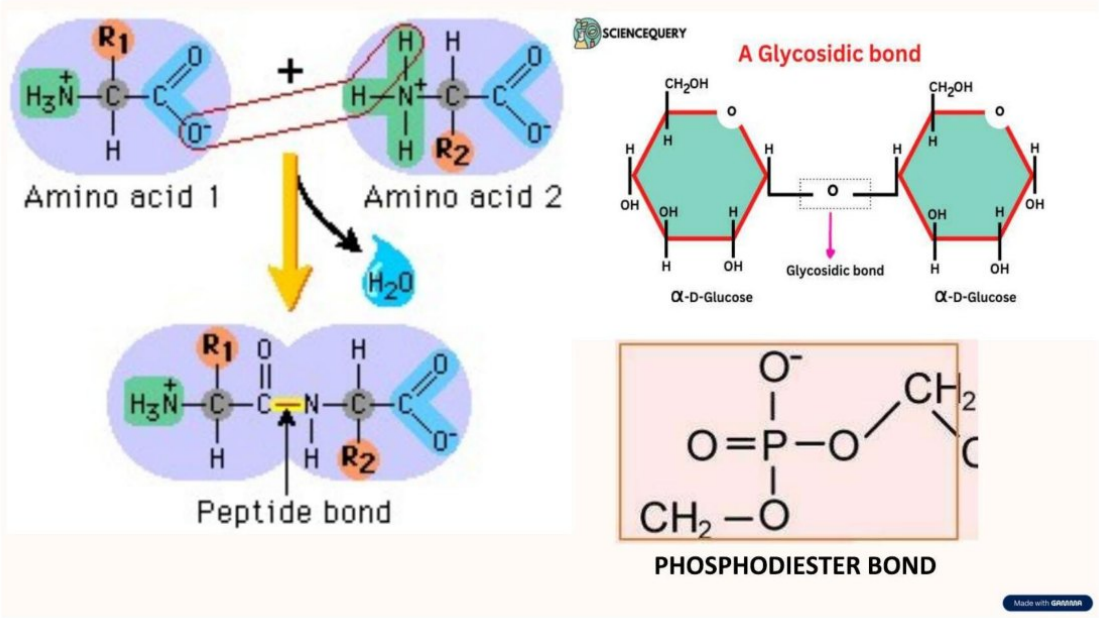


Fig. 10 — Fig. B11 — Peptide bond formation between two amino acids (dehydration condensation, releasing H2O) *(bank p. 12)*

Answer. The primary structure of a peptide is its linear sequence of amino acids, read from the free N-terminus (amino end) to the free C-terminus (carboxyl end), with successive residues joined by peptide bonds (formed by dehydration condensation between the -COOH of one residue and the -NH2 of the next). FYP = Phe – Tyr – Pro → H2N–Phenylalanine–Tyrosine–Proline–COOH (tripeptide, 2 peptide bonds) • ATF = Ala – Thr – Phe → H2N–Alanine–Threonine–Phenylalanine–COOH (tripeptide, 2 peptide • bonds) DCP = Asp – Cys – Pro → H2N–Aspartic acid–Cysteine–Proline–COOH (tripeptide, 2 peptide bonds) •

QB13. Describe the structure of the DNA double helix (Watson–Crick model) and differentiate DNA

7 marks · from the Unit 1 bank

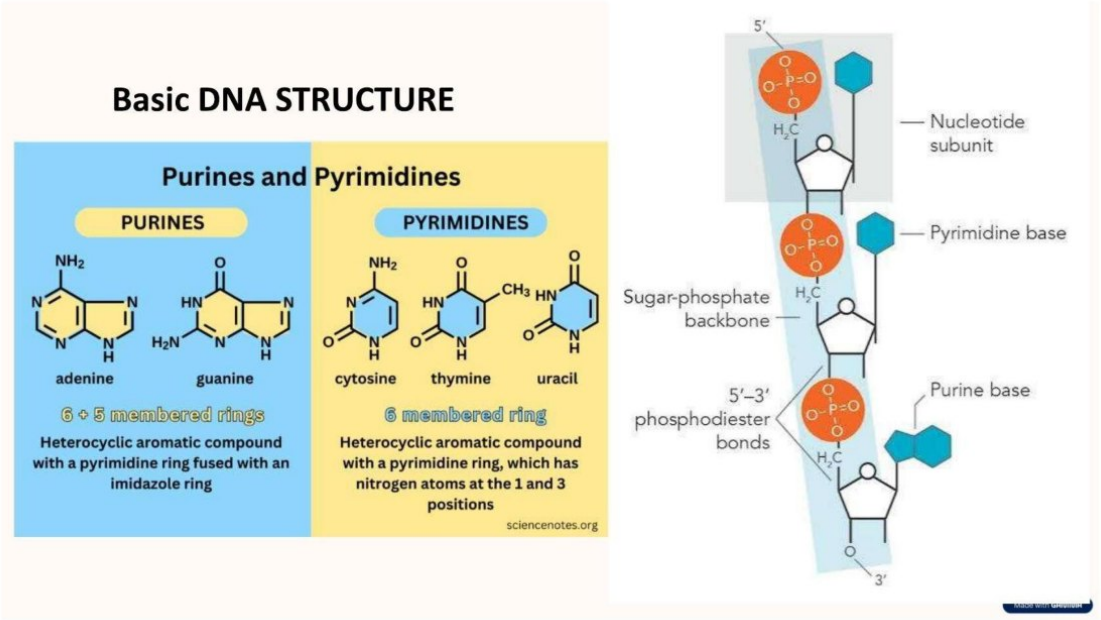


Fig. 11 — Fig. B13a — Purine and pyrimidine bases; the nucleotide sugar-phosphate backbone (bank p. 13)

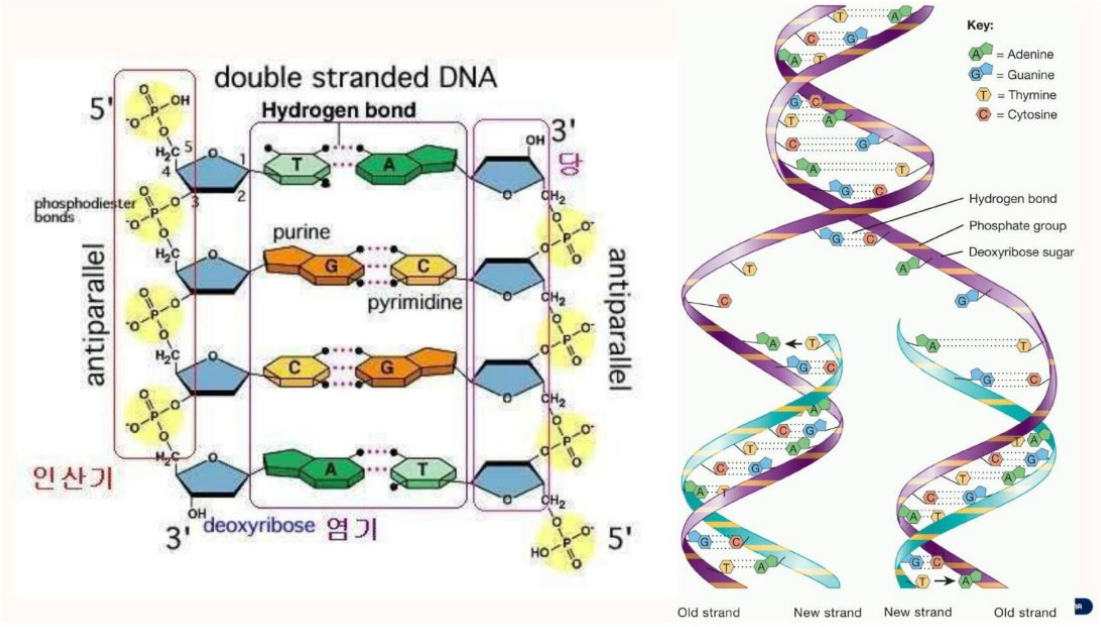


Fig. 12 — Fig. B13b — The DNA double helix showing antiparallel strands, base pairing and replication (bank p. 13)

Answer. from RNA. Ans: James Watson and Francis Crick proposed the double helix model of DNA in 1953, using X-ray diffraction data obtained by Rosalind Franklin and Maurice Wilkins. Key structural features: A nucleotide (the monomer) consists of a nitrogenous base (purine — adenine/guanine; or pyrimidine • — cytosine/thymine), a pentose sugar (deoxyribose), and a phosphate group. Two antiparallel polynucleotide strands wind into a right-handed helix; the sugar-phosphate backbone • (linked by 3'-5' phosphodiester bonds) lies on the outside, with nitrogenous bases pointing inward. Complementary base pairing (Chargaff's rule): Adenine pairs with Thymine via 2 hydrogen bonds; • Guanine pairs with Cytosine via 3 hydrogen bonds. The helix has a pitch of 34 Å per turn (10 base pairs/turn) and displays a major groove and a

minor • groove, which are binding sites for regulatory proteins. DNA vs. RNA: Feature DNA RNA Sugar Deoxyribose (no 2'-OH) Ribose (has 2'-OH) Bases A, T, G, C A, U, G, C (U replaces T) Strands Usually double-stranded Usually single-stranded Stability Very stable (no 2'-OH) Less stable (2'-OH promotes hydrolysis) Function Genetic information storage Gene expression (mRNA, tRNA, rRNA)

UNIT 2 · Cells and Biomolecules — Function

7 diagram-based questions · 8 diagrams in this unit

QQ1. What are enzymes? Discuss their structure, characteristics, and significance as biocatalysts in biological processes

7 marks · from the Unit 2 bank

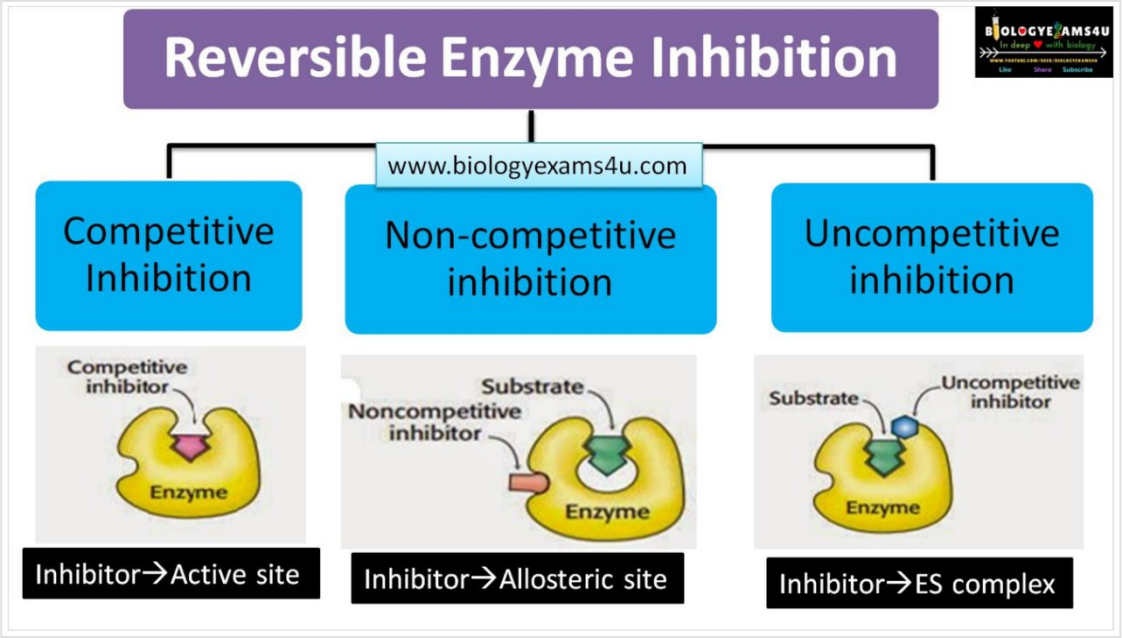


Fig. 13 — Fig: Reversible modes of enzyme inhibition — competitive, non-competitive, and uncompetitive. (bank p. 5)

Answer. Definition: Enzymes are biological macromolecules — mostly proteins, though some are RNAs (ribozymes) — that act as biocatalysts, accelerating biochemical reactions without being consumed in the process. Characteristics of Enzymes: • Specificity: Highly specific for their substrates, explained by the lock-and-key or induced-fit models. • Catalytic Efficiency: Accelerate reactions by lowering the activation energy. • Reusability: Not consumed during the reaction and can be used repeatedly. • Optimal Conditions: Function best under specific pH, temperature, and ionic strength. • Regulation: Activity can be modulated by inhibitors, activators, or feedback mechanisms. Mechanism of Action: 1. The substrate binds to the enzyme's active site, forming an enzyme-substrate complex. 2. The enzyme stabilizes the transition state, reducing the activation energy. 3. Products are released, and the enzyme is free to catalyze another reaction. Types of Enzymes (based on reaction catalyzed): Oxidoreductases, Transferases, Hydrolases, Lyases, Isomerases, and Ligases. Enzyme Inhibition: Competitive (inhibitor competes at the active site), Non-competitive (inhibitor binds an allosteric site), and Uncompetitive (inhibitor binds only the enzyme-substrate complex). Significance: Enzymes are essential for life processes such as metabolism, DNA replication, and energy production, and serve as efficient, eco-friendly alternatives to chemical catalysts in medicine, industry (food, textiles, biofuels), and research (e.g., DNA polymerase in PCR).

QQ2. Define fermentation. Discuss the different steps of fermentation

7 marks · from the Unit 2 bank

making, and baking. `Glucose → 2 Ethanol + 2 CO2 + Energy (ATP)`

◦ **Lactic Acid Fermentation (e.g., muscle cells, some bacteria):** Pyruvate is reduced to lactic acid, regenerating NAD^+. Occurs in oxygen-deprived muscles (causing the "burn") and in dairy fermentation.

`Glucose → 2 Lactic Acid + Energy (ATP)`

4. **Byproducts:** Ethanol and CO_2 (alcoholic fermentation) or lactic acid (lactic acid fermentation).

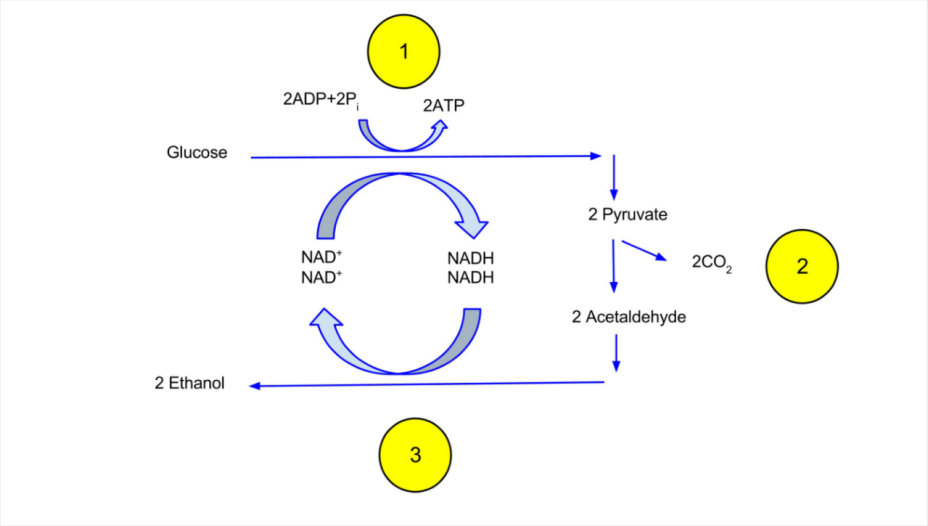


Fig. 14 — Fig: Pathway of alcoholic fermentation showing glycolysis, ATP/NADH cycling, and ethanol formation. *(bank p. 6)*

Answer. Definition: Fermentation is a metabolic process in which microorganisms like bacteria, yeast, or molds convert sugars and other organic compounds into simpler molecules, often producing energy, under anaerobic (oxygen-free) conditions. It is widely used in food and beverage production and various industrial processes. Steps of Fermentation: 1. Sugar Breakdown (Glycolysis): Glucose is split into two molecules of pyruvate, producing a small amount of ATP and NADH. Regeneration of NAD+: Since glycolysis requires NAD+ to continue, and oxygen is absent, the cell regenerates NAD+ through 2. fermentation. 3. Fermentation Pathways: Alcoholic Fermentation (e.g., Yeast): Pyruvate is converted into ethanol and CO2, regenerating NAD+. Used in brewing, wine- ◦ making, and baking. Glucose → 2 Ethanol + 2 CO2 + Energy (ATP) Lactic Acid Fermentation (e.g., muscle cells, some bacteria): Pyruvate is reduced to lactic acid, regenerating NAD+. Occurs in ◦ oxygen-deprived muscles (causing the "burn") and in dairy fermentation. Glucose → 2 Lactic Acid + Energy (ATP) 4. Byproducts: Ethanol and CO2 (alcoholic fermentation) or lactic acid (lactic acid fermentation). Applications: Bread making (CO2 causes dough to rise), alcoholic beverages, dairy products (yogurt, cheese), and industrial production of biofuels, antibiotics, and other chemicals.

QQ3. Define Bioenergy. What are the different forms of Bioenergy? Write down the different production processes and challenges of

2 marks · from the Unit 2 bank

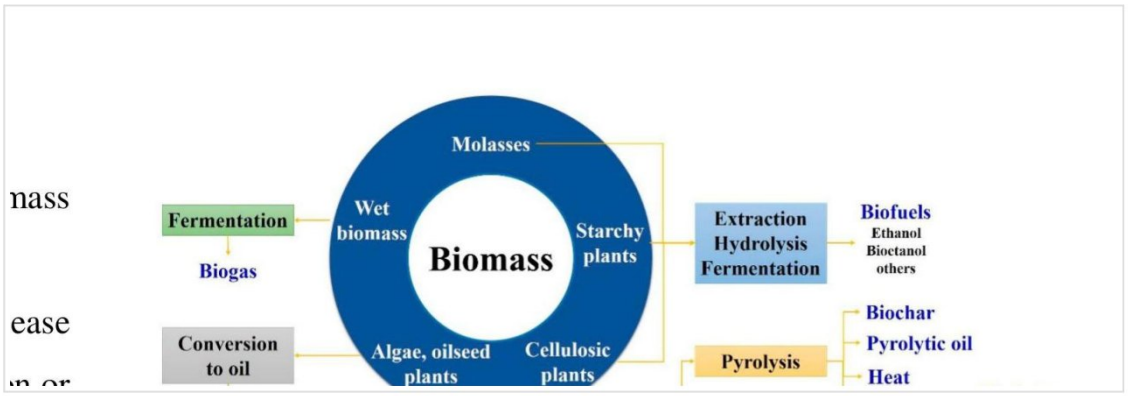


Fig. 15 — Fig: Biomass conversion routes into biofuels, biochar, pyrolytic oil, heat, and syngas. *(bank p. 7)*

Answer. Bioenergy. [7] Definition: Bioenergy is energy derived from biological sources such as plants, animals, and microorganisms (biomass). Unlike fossil fuels, it is renewable and can be replenished over time. Forms of Bioenergy: • Biogas: Produced by anaerobic decomposition of organic matter; mainly methane (CH4), used for cooking, heating, or electricity. • Biofuels: Liquid fuels for transportation — Ethanol (from fermenting sugars in crops) and Biodiesel (from plant oils or animal fats). • Biomass for Electricity and Heat: Direct burning of biomass to generate power through steam turbines. • Biohydrogen: Hydrogen gas produced from organic material, explored for fuel cells. Production Processes: • Combustion: Direct burning of biomass to release heat. • Anaerobic Digestion: Breakdown of organic matter without oxygen to produce biogas. • Fermentation: Conversion of crop sugars into ethanol by microorganisms. • Transesterification: Converts vegetable oils/animal fats into biodiesel. • Gasification: Biomass heated in low oxygen to produce syngas (H2, CO, CH4). • Pyrolysis: Biomass heated in absence of oxygen, yielding bio-oil, charcoal, and syngas. Challenges of Bioenergy: • Land Use and Food Security: Crops grown for biofuel may compete with food production. • Deforestation: Unsustainable harvesting of forest biomass causes biodiversity loss. • Energy Efficiency: Some methods (e.g., corn ethanol) have a low net energy gain. • Carbon Emissions: The full lifecycle of production can still generate emissions. • Water Use: Bioenergy crops often need large amounts of water. • Cost: Production can be expensive and may require government subsidies.

QQ4. What is photosynthesis? Discuss the different steps involved in photosynthesis

7 marks · from the Unit 2 bank

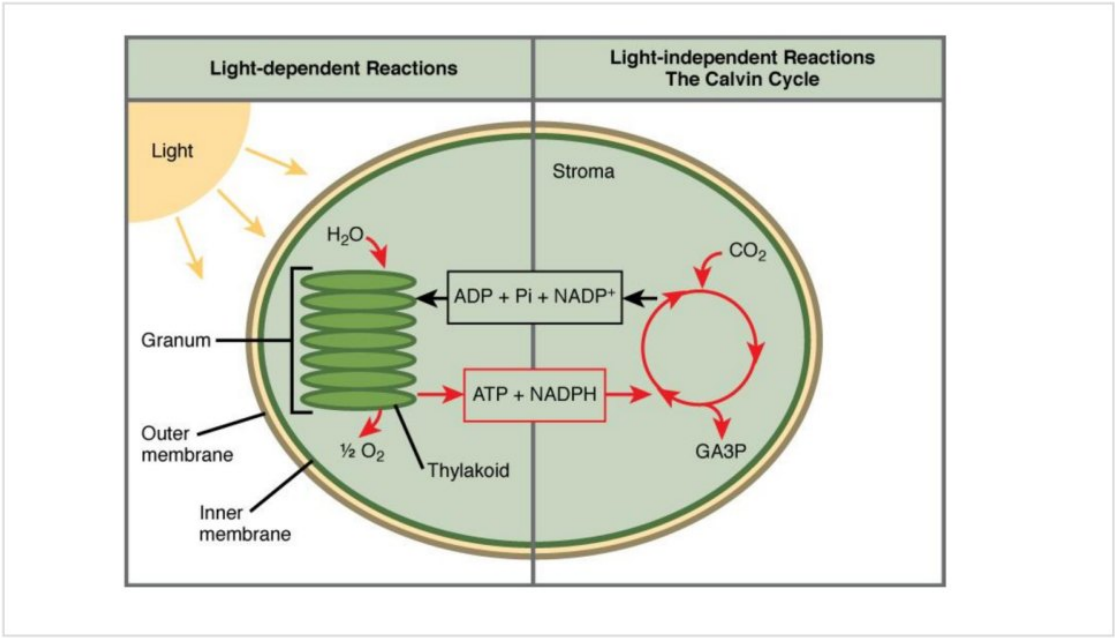


Fig. 16 — Fig: Chloroplast structure showing the light-dependent reactions (thylakoid) and the light-independent Calvin Cycle (stroma). *(bank p. 8)*

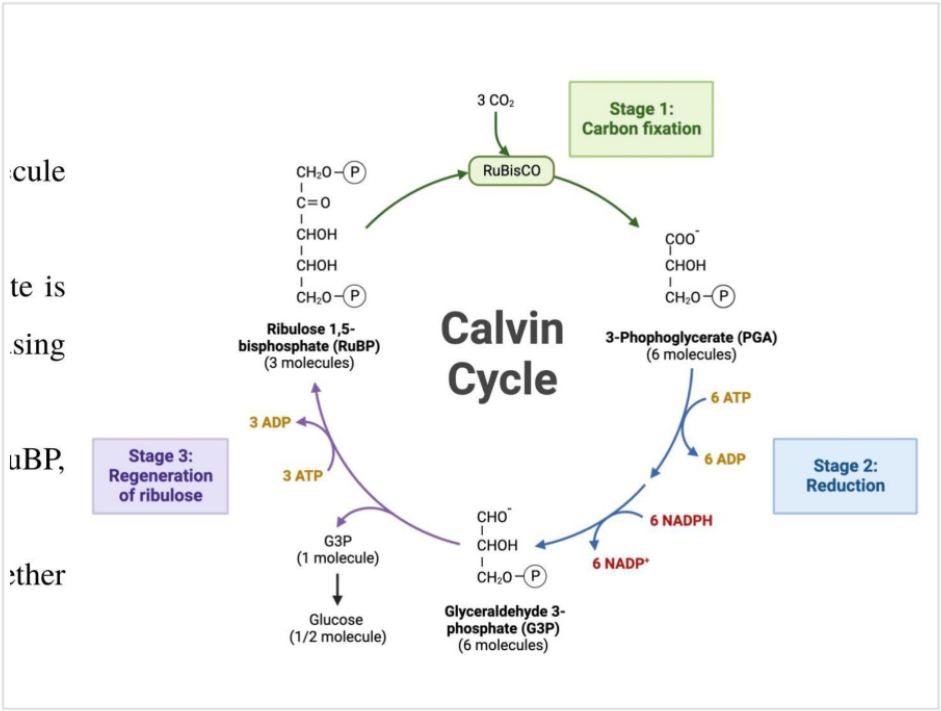


Fig. 17 — Fig: The Calvin Cycle — carbon fixation, reduction, and regeneration of ribulose. *(bank p. 9)*

Answer. Definition: Photosynthesis is the process by which green plants, algae, and certain bacteria convert light energy into chemical energy, storing it in the form of glucose. It occurs mainly in the chloroplasts, where chlorophyll absorbs sunlight. 6 CO2 + 6 H2O + light energy → C6H12O6 + 6 O2 1. Light-Dependent Reactions (occur in the thylakoid membrane): 1. Light absorption: Pigments (chlorophyll, carotenoids) absorb light. 2. Excitation of electrons: Light energy excites electrons, transferring them to electron acceptors. 3. Transfer of electrons: Electrons pass through a series of carriers, forming a proton gradient. 4. ATP synthesis: The

proton gradient drives ATP production through chemiosmosis; water is split, releasing O2. 2. Light-Independent Reactions (Calvin Cycle) (occur in the stroma): 1. Carbon fixation: CO2 is fixed into a 3-carbon molecule (3-phosphoglycerate) by the enzyme RuBisCO. 2. Reduction: 3-phosphoglycerate is reduced to glyceraldehyde-3-phosphate (G3P) using ATP and NADPH. 3. Regeneration of RuBP: G3P regenerates ribulose-1,5-bisphosphate (RuBP) to continue the cycle. 4. Glucose production: G3P molecules combine to form glucose. Importance of Photosynthesis: Produces organic compounds that support life on Earth, releases oxygen into the atmosphere, and forms the basis of most food chains.

QQ6. What are pesticides? Write down four advantages and four disadvantages of pesticides

7 marks · from the Unit 2 bank

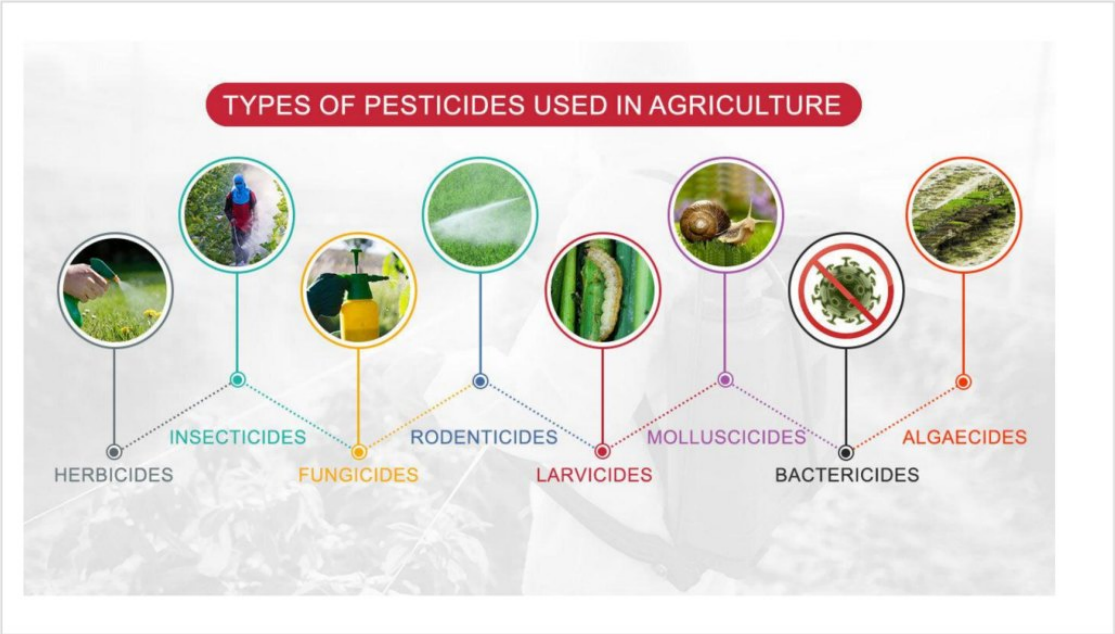


Fig. 18 — Fig: Major types of pesticides used in agriculture. *(bank p. 10)*

Answer. Definition: Pesticides are chemicals or biological agents used to control pests — insects, weeds, fungi, and other organisms — that damage crops, livestock, or structures. Types of Pesticides: • Insecticides (e.g., chlorpyrifos, permethrin) — control insect pests. • Herbicides (e.g., glyphosate, 2,4-D) — control unwanted weeds. • Fungicides (e.g., copper sulfate) — control fungal diseases. • Rodenticides (e.g., warfarin) — control rodent populations. • Bactericides and Nematicides — control harmful bacteria and nematodes respectively. Four Advantages: 1. Increased Crop Yield: Controls pests, ensuring healthier crops and reduced damage. 2. Protection Against Disease: Protects plants and animals from disease-causing organisms. 3. Economic Benefits: Higher yields and reduced crop loss improve farmers' profits. 4. Faster Action: Acts quickly to control pest outbreaks. Four Disadvantages: 1. Environmental Pollution: Runoff can contaminate soil, water, and air. 2. Non-target Effects: Can harm beneficial organisms like bees, birds, and aquatic life. 3. Human Health Risks: Toxicity from exposure, and long-term effects like cancer or reproductive issues. 4. Resistance & Biodiversity Loss: Overuse leads to pest resistance and reduced biodiversity.

QQ8. What is biological pest control? Discuss its types with examples, and mention its applications

7 marks · from the Unit 2 bank

Biological Pest Control

Q8. What is biological pest control? Discuss its types with examples, and mention its applications. [7]

Definition: Biological pest control is an environmentally friendly, sustainable approach to managing pests parasites, or pathogens, instead of synthetic chemicals like pesticides. It works by exploiting natural relationships

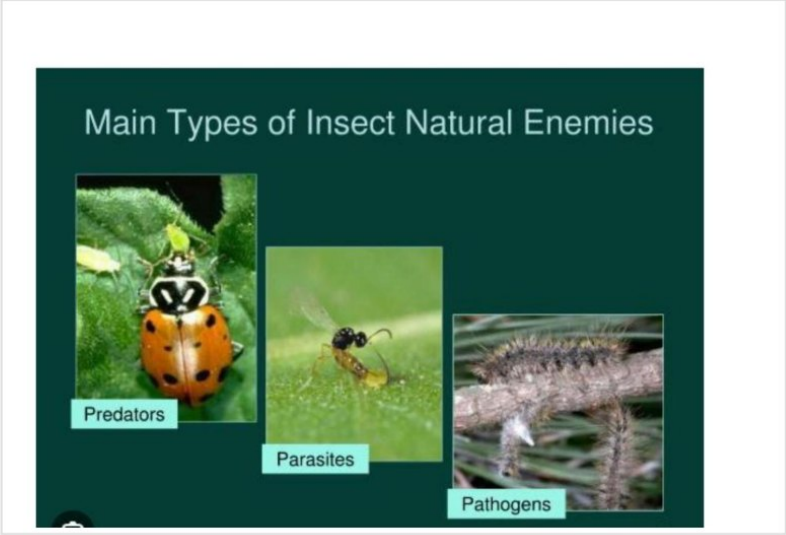


Fig. 19 — Fig: Main types of insect natural enemies — predators, parasites, and pathogens. *(bank p. 12)*

Answer. Definition: Biological pest control is an environmentally friendly, sustainable approach to managing pests using natural predators, parasites, or pathogens, instead of synthetic chemicals like pesticides. It works by exploiting natural relationships between organisms. Types of Biological Pest Control: • Predators: Organisms that hunt and kill pests for food. Examples: Ladybugs (prey on aphids), Lacewing larvae (consume aphids, mealybugs), Praying mantises (eat a wide variety of insects). • Parasites: Live in or on a host organism, ultimately killing it to complete their life cycle. Examples: Parasitic (Trichogramma) wasps (lay eggs inside pest insects), Nematodes (parasitize soil-dwelling pests like grubs). • Pathogens: Microorganisms (bacteria, fungi) that infect and kill pests. Examples: Bacillus thuringiensis (Bt) — toxic to caterpillars; Beauveria bassiana — a fungus that infects termites, whiteflies, and beetles. Applications of Biological Control: • Agriculture: Controls pests on vegetables, fruits, and field crops. • Horticulture & Gardening: Home gardeners use predators like ladybugs. • Greenhouses: Used in enclosed spaces to control whiteflies and aphids. • Forestry: Manages pests threatening forests, such as the pine beetle.

QQ9. Define Vaccines. How do Vaccines work? Write down the importance of vaccines

7 marks · from the Unit 2 bank

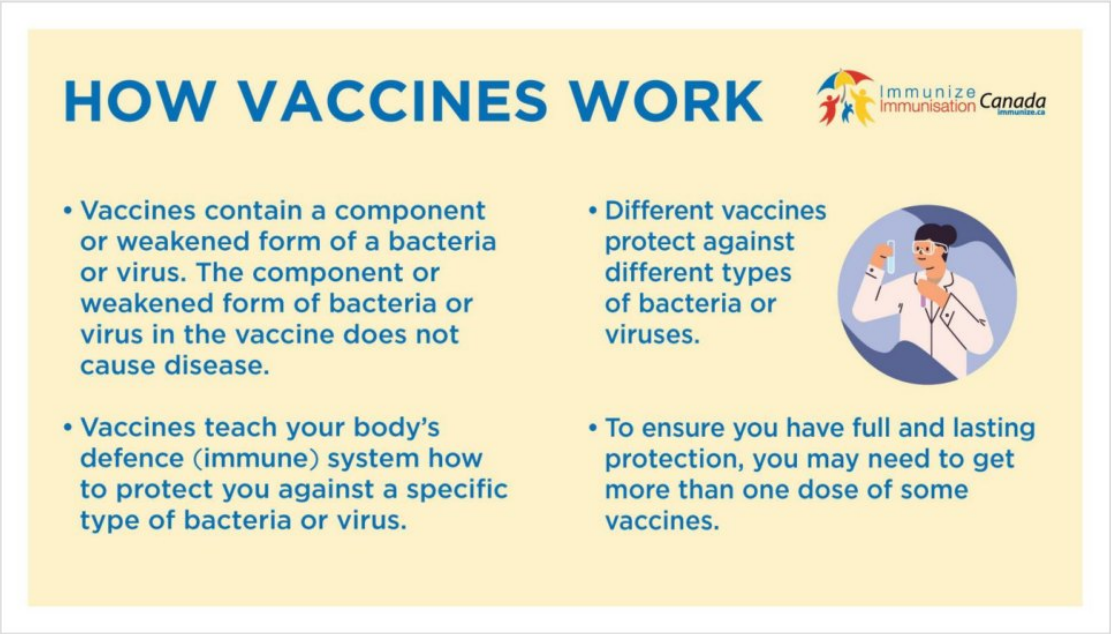


Fig. 20 — Fig: How vaccines work — training the immune system without causing disease. *(bank p. 13)*

Answer. Definition: Vaccines are biological preparations designed to provide immunity to specific diseases. They stimulate the immune system to recognize and fight pathogens (bacteria, viruses, or parasites) without causing the disease itself. How Vaccines Work: • Immune System Activation: Vaccines contain weakened or inactivated pathogens (or parts of them) that trigger an immune response without causing illness. • Antigen Exposure: Antigens in the vaccine cause the immune system to produce antibodies that can neutralize the pathogen. • Memory Cells: The immune system produces memory B-cells and T-cells that "remember" the pathogen, enabling a faster, stronger response on future exposure. Types of Vaccines: RNA, DNA, Viral vector, Virus-like particle, Protein sub-unit, and Inactivated virus vaccines. Importance of Vaccines: • Prevention of Disease: Prevent the spread of infectious diseases, reducing illness in the population. • Herd Immunity: Protects those who cannot be vaccinated (infants, immunocompromised, elderly) when a large portion of the population is vaccinated. • Eradication of Diseases: Led to the eradication of smallpox and near-elimination of polio. • Reduced Healthcare Costs: Prevents hospitalizations and long-term care needs. Challenges in Vaccination: Vaccine hesitancy due to misinformation, limited access in underdeveloped regions, and difficulty developing vaccines for rapidly mutating pathogens (e.g., HIV, influenza).

UNIT 3 · Structure and Function of Ecosystem

10 diagram-based questions · 12 diagrams in this unit

QQ11. What is food chain? Explain in brief with schematic diagram

2 marks · from the Unit 3 bank



Fig. 21 — Fig: A simple food chain — producer to final consumer/decomposer. *(bank p. 2)*

Answer. Answer: A food chain is the straight chain of organisms that starts with the producer, through which energy is transferred to the carnivores, and ends with the decomposer in the food web. It explains the feeding relationship of organisms — producers (autotrophs) → primary consumers (herbivores) → secondary/tertiary consumers (carnivores) → decomposers.

QQ1. Discuss the biotic components of the ecosystem and their organization

7 marks · from the Unit 3 bank

Community: Populations of different species living together and interacting, forming food webs; communities occur in habitats determined by topography, vegetation structure, geology and the surrounding medium.

Ecosystem: Encompasses both the living (biotic) and non-living (abiotic) components of an area, with strong interaction between them; major processes are nutrient cycling and energy flow.

Biome: A group of organisms sharing common characteristics correlated with the environment they exist in; may span more than one continent (e.g., tropical rainforest, coniferous forest/taiga, grassland, desert scrub).

Biosphere: The highest organizational level — that part of the Earth and atmosphere in which life exists, including the surface layer of land, oceans, sediments and the part of the atmosphere occupied by life.

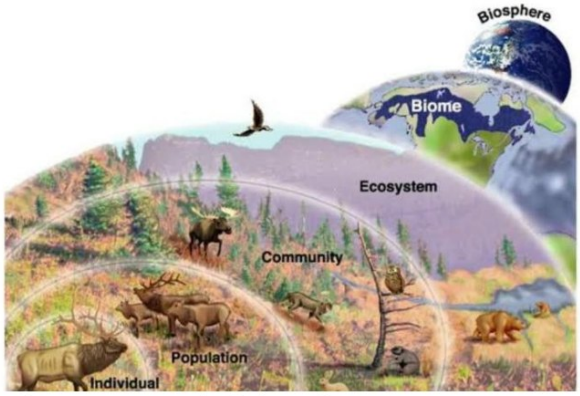


Fig. 22 — Fig: Levels of biological organization — individual, population, community, ecosystem, biome and biosphere. (bank p. 4)

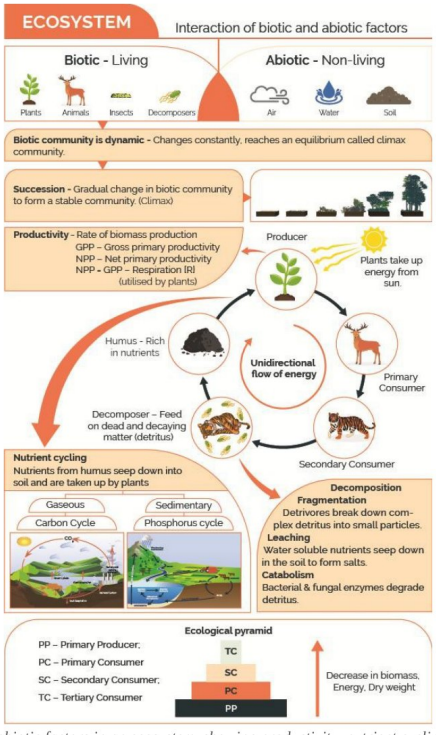


Fig. 23 — Fig: Interaction of biotic and abiotic factors in an ecosystem, showing productivity, nutrient cycling and the ecological pyramid. (bank p. 4)

Answer. Answer: Biotic components are the living parts of an ecosystem. They include producers, consumers and decomposers: Producers: Autotrophic, chlorophyll-bearing organisms such as plants, which produce their own food ● through photosynthesis. Primary Consumers: Herbivores that get nutrition by eating plants directly, e.g. rabbit, deer and cow. ● Secondary Consumers: Carnivores that eat the flesh of herbivores, e.g. cats and dogs. ● Tertiary

Consumers: Carnivores that prey upon other carnivores, e.g. lion, tiger and vulture. ● Decomposers: Attack the dead remains of producers and consumers and degrade complex organic ● substances into simpler compounds, playing a vital role in maintaining the dynamic nature of the ecosystem. Organization within the biotic component follows a defined hierarchy: Individual: All individuals of a species potentially able to interbreed and produce fertile offspring (e.g., ● the salmon Salmo salar). Population: A group of individuals of the same species living in a particular area at the same time; each ● population is genetically distinct to some degree and has a size, birth rate, death rate and growth rate. Community: Populations of different species living together and interacting, forming food webs; ● communities occur in habitats determined by topography, vegetation structure, geology and the surrounding medium. Ecosystem: Encompasses both the living (biotic) and non-living (abiotic) components of an area, with ● strong interaction between them; major processes are nutrient cycling and energy flow. Biome: A group of organisms sharing common characteristics correlated with the environment they exist ● in; may span more than one continent (e.g., tropical rainforest, coniferous forest/taiga, grassland, desert scrub). Biosphere: The highest organizational level — that part of the Earth and atmosphere in which life exists, ● including the surface layer of land, oceans, sediments and the part of the atmosphere occupied by life.

QQ3. What are trophic levels? Discuss each trophic level with example

7 marks · from the Unit 3 bank

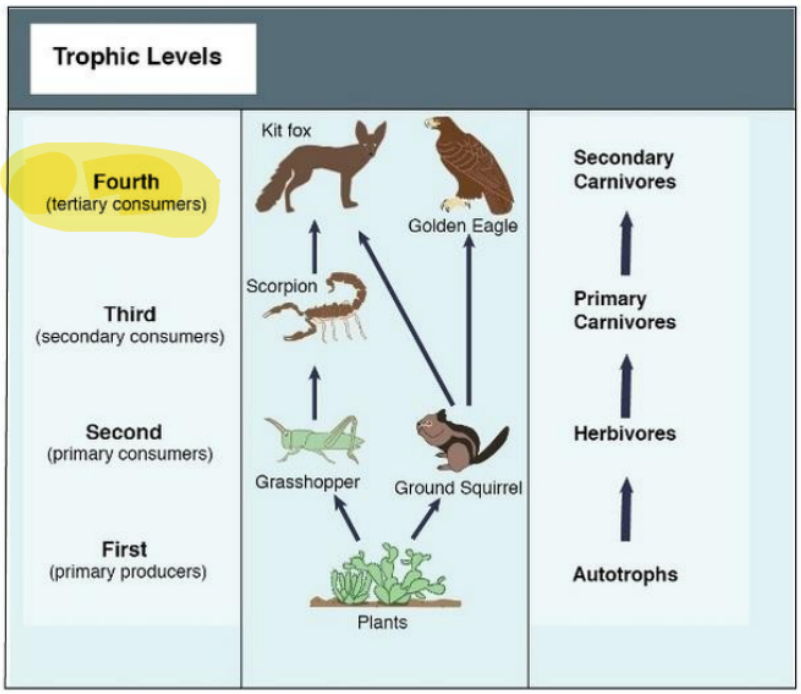


Fig. 24 — Fig: Trophic levels and food web — from primary producers to tertiary consumers. *(bank p. 6)*

Answer. Answer: A trophic level refers to the position an organism occupies in a food chain, based on its feeding relationships. Energy and biomass decrease progressively from one trophic level to the next. First Trophic Level — Primary Producers: Autotrophs / green plants that convert solar energy into ● chemical energy through photosynthesis, forming the base of the food chain. Example: grass, plants. Second Trophic Level — Primary Consumers: Herbivores that feed directly on producers. Example: ● grasshopper, ground squirrel, rabbit, deer, cow. Third Trophic Level — Secondary Consumers: Primary carnivores that feed on herbivores. Example: ● frog, scorpion, cats and dogs. Fourth Trophic Level — Tertiary Consumers: Secondary carnivores that prey upon other carnivores; ● these are typically the top of the food chain. Example: snake, kit fox, golden eagle, hawk, lion, tiger. Decomposers: Fungi and bacteria

that break down dead organisms and waste from all trophic levels, ● returning nutrients to the soil. Example: fungi, earthworm. At each successive trophic level, only about 10% of the energy absorbed by one level is passed on to the next (the rest is lost as heat via respiration), which is why food chains are typically limited to 4–5 levels and why the biomass/energy pyramid narrows towards the top.

QQ4. How does energy flow in ecosystem? Discuss elaborately

7 marks · from the Unit 3 bank

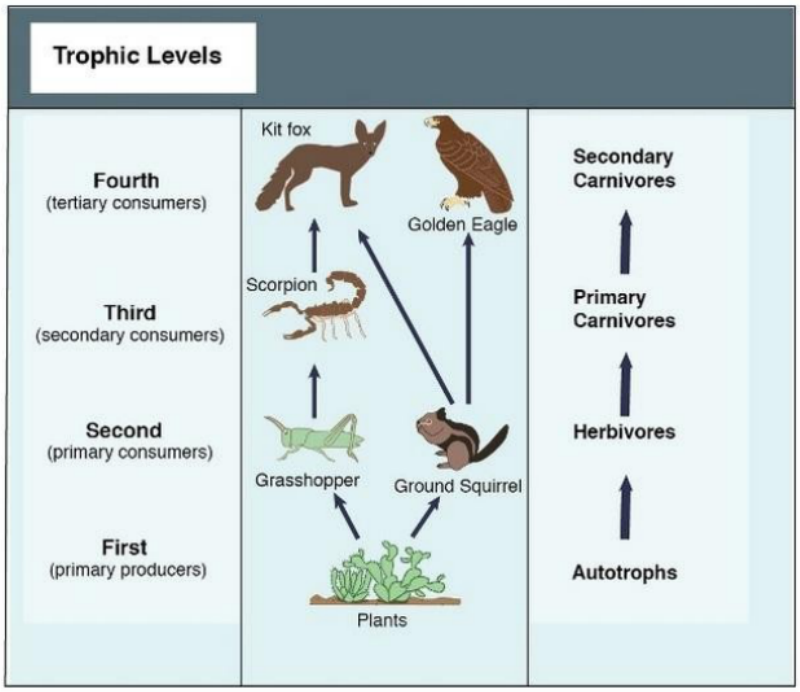


Fig. 25 — Fig: Energy transfer and loss across trophic levels (energy pyramid). *(bank p. 6)*

Answer. Answer: Ecological energetics is the study of the movement of energy through an ecosystem. The sun provides all the energy that living beings require. Green plants absorb around 1% of total radiant ● energy for distribution across the ecosystem — despite its small size, this is adequate to maintain all life on Earth. Fuel enters the food chain at the level of producers in the form of solar energy. Plants convert solar energy ● into chemical energy through photosynthesis. This chemical energy is then passed from one trophic level to the next along a food chain. First Law of Thermodynamics: energy can neither be created nor destroyed — solar energy is only ● transferred from one system/form to another. Second Law of Thermodynamics: during every transformation, a portion of energy is dissipated into the ● surroundings as heat — hence energy flow is unidirectional, unlike nutrient cycling. As shown in the energy pyramid, roughly 10% of the energy at one trophic level is transferred to the next (Primary Producers ≈10,000 Kcal/m²/year → Primary Consumers ≈1,000 → Secondary Consumers ≈100 → Tertiary Consumers ≈10 Kcal/m²/year), with the majority lost as heat through cellular respiration at every step. Decomposers live off detritus, decomposing waste and dead matter from every trophic level.

QQ5. Discuss the carbon cycle in details

7 marks · from the Unit 3 bank

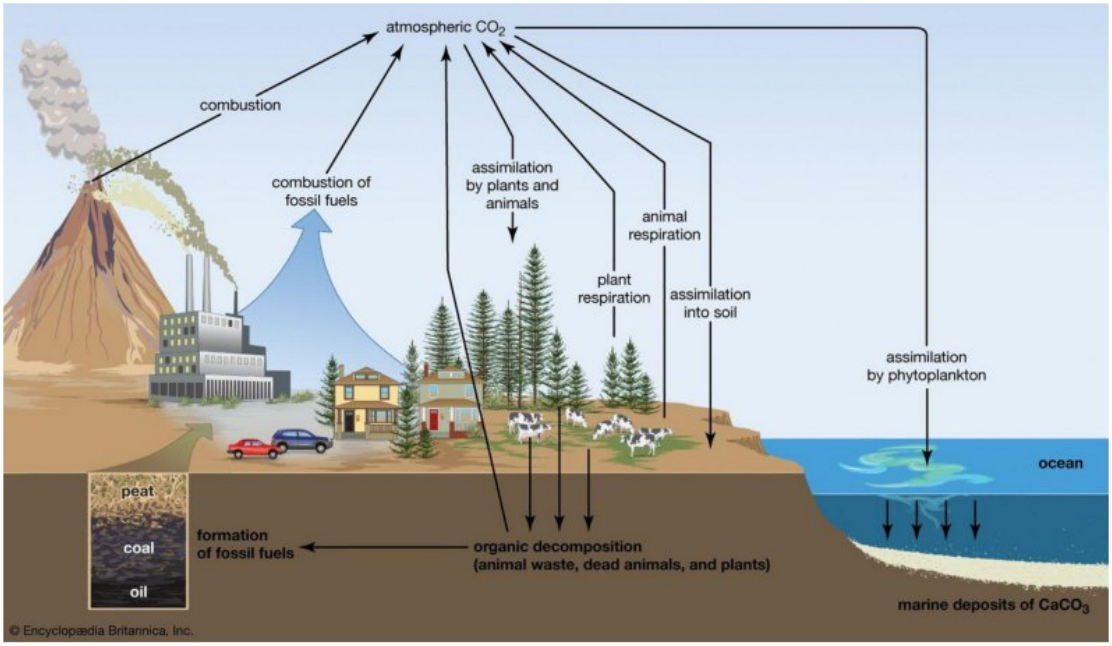


Fig. 26 — Fig: The Carbon Cycle. *(bank p. 7)*

Answer. Answer: The carbon cycle is the part of the biogeochemical cycle by which carbon is exchanged among the biosphere, pedosphere, geosphere, hydrosphere and atmosphere of Earth. Carbon is the main component of biological compounds as well as a major component of many minerals. ● The cycle comprises a sequence of events key to making Earth capable of sustaining life, describing the movement of carbon as it is recycled and reused, as well as long-term carbon sequestration and release from carbon sinks. Key processes: CO2 is fixed by plants and phytoplankton through photosynthesis (assimilation) and ● returned to the atmosphere through plant and animal respiration, combustion (of fossil fuels and biomass) and volcanic activity. Dead animals, plants and waste undergo organic decomposition; over geological time, this can lead to the ● formation of fossil fuels (peat, coal, oil). CO2 dissolves directly from the atmosphere into oceans/lakes and in precipitation, forming carbonic acid ● (contributing to ocean acidity) — it may also be absorbed by rocks through weathering, or washed into the ocean as marine deposits of CaCO3 (via assimilation by phytoplankton and marine sedimentation). Carbon exists mainly as CO2 and methane (CH4) in the atmosphere — both absorb and retain heat, ● contributing to the greenhouse effect. Methane has a larger greenhouse effect per volume than CO2 but exists in lower concentrations and is shorter-lived, so CO2 contributes more to the overall global greenhouse effect. Human impact: Human activities (land-use change, mining/combustion of fossil fuels, cement ● manufacture) have disturbed the fast carbon cycle for centuries. Atmospheric CO2 had increased nearly 52% over pre-industrial levels by 2020, forcing greater atmospheric/surface heating and reducing ocean pH — fundamentally altering marine chemistry and contributing to human-caused climate change.

QQ6. Describe the processes involved in nitrogen cycle

7 marks · from the Unit 3 bank

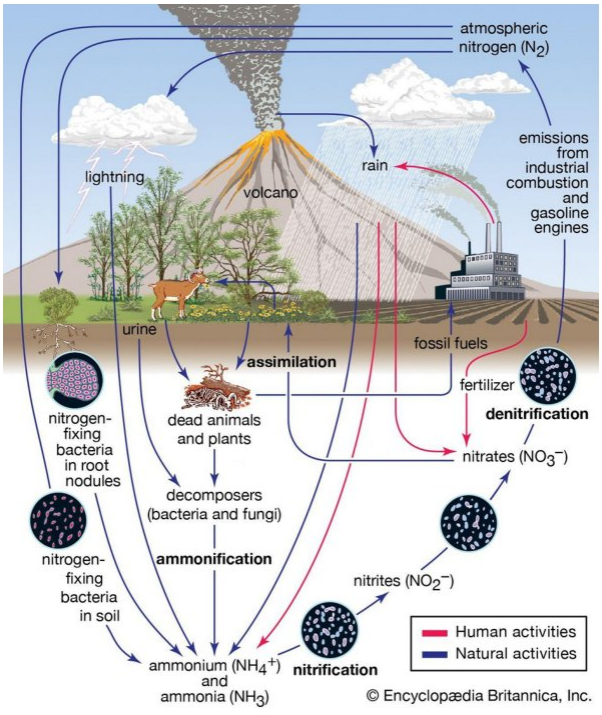


Fig. 27 — Fig: The Nitrogen Cycle — fixation, ammonification, nitrification, assimilation and denitrification. (bank p. 8)

Answer. Answer: The nitrogen cycle is the biogeochemical cycle by which nitrogen is converted into multiple chemical forms as it circulates among atmospheric, terrestrial and marine ecosystems. Although 78% of the atmosphere is N2, this form has limited biological availability, leading to scarcity of usable nitrogen. Nitrogen Fixation: Conversion of inert atmospheric N2 into ammonia/ammonium by nitrogen-fixing ● bacteria (e.g., Rhizobium in root nodules, free-living bacteria in soil), lightning, or industrial fertilizer production. Ammonification: Decomposers (bacteria and fungi) break down dead plants/animals and nitrogenous ● waste (urine), releasing ammonium (NH4+)/ammonia (NH3) into the soil. Nitrification: Nitrifying bacteria oxidize ammonium to nitrite (NO2−) and then to nitrate (NO3−). ● Assimilation: Plants take up ammonium and nitrate from the soil and incorporate them into organic ● molecules such as proteins; animals assimilate nitrogen by consuming plants. Denitrification: Denitrifying bacteria convert nitrate back into atmospheric N2, completing the cycle. ● Human activities — fossil fuel combustion, use of artificial nitrogen fertilizers and release of nitrogen in wastewater — have dramatically altered the global nitrogen cycle, which can negatively affect both the natural environment and human health. Since nitrogen availability affects primary production and decomposition rates, the nitrogen cycle is of particular interest to ecologists. ENV

QQ8. Discuss the Phosphorus and Sulfur cycle

7 marks · from the Unit 3 bank

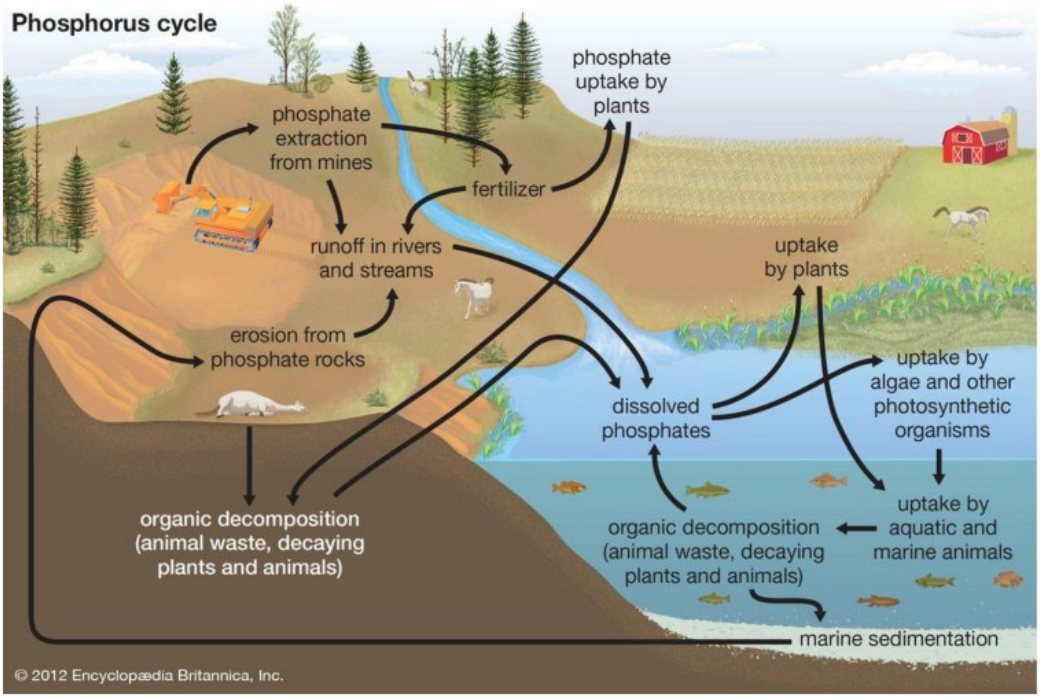


Fig. 28 — Fig: The Phosphorus Cycle. *(bank p. 9)*

Sulphur enters the atmosphere as SO2 and H2S from natural sources (volcanic activi
living things and waste products) and anthropogenic sources (smelting and burning c

Sulphur compounds return to Earth as wet deposition (acid rain, snow) or dry deposi
deposition; sulfates accumulate in soil (SO_4^{2-}) and in water.

Microorganisms in soil and water convert sulfur compounds between forms; sulfates
deposited in sediments, and iron sulfides accumulate in sediment, eventually forming
geological time.

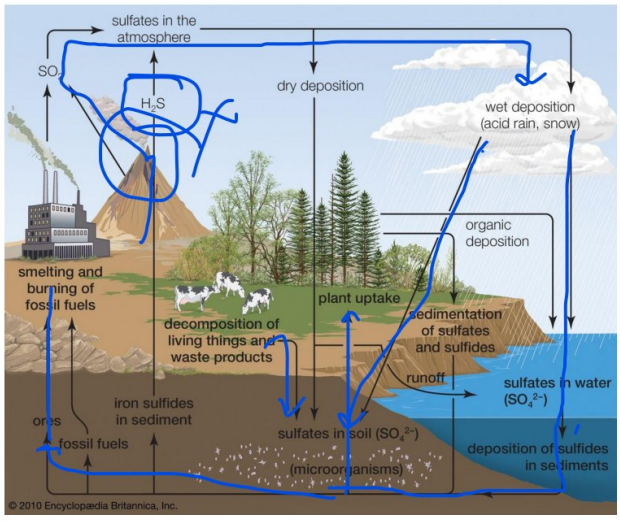


Fig. 29 — Fig: The Sulphur Cycle. *(bank p. 9)*

Answer. Phosphorus Cycle Phosphorus is released from phosphate rocks through weathering and erosion, and by mining/extraction; it ● moves as runoff into rivers and streams. Phosphate is taken up by plants directly from soil or applied as fertilizer; it also dissolves in water bodies ● as dissolved phosphates, taken up by algae and other photosynthetic/aquatic organisms. Organic decomposition of animal waste and decaying plants/animals (on land and in water) returns ● phosphorus to the soil/water; in water bodies, unused phosphorus settles as marine sedimentation, eventually reforming phosphate rock over geological time. Unlike carbon or

nitrogen, phosphorus has no significant atmospheric/gaseous phase — it is therefore ● called a sedimentary cycle, and weathering of rock is its principal natural source. Sulphur Cycle Sulphur enters the atmosphere as SO2 and H2S from natural sources (volcanic activity, decomposition of ● living things and waste products) and anthropogenic sources (smelting and burning of fossil fuels). Sulphur compounds return to Earth as wet deposition (acid rain, snow) or dry deposition, and as organic ● deposition; sulfates accumulate in soil ($SO4^{2-}$) and in water. Microorganisms in soil and water convert sulfur compounds between forms; sulfates and sulfides may be ● deposited in sediments, and iron sulfides accumulate in sediment, eventually forming ores/fossil fuels over geological time.

QQ9. Describe the Oxygen cycle

7 marks · from the Unit 3 bank

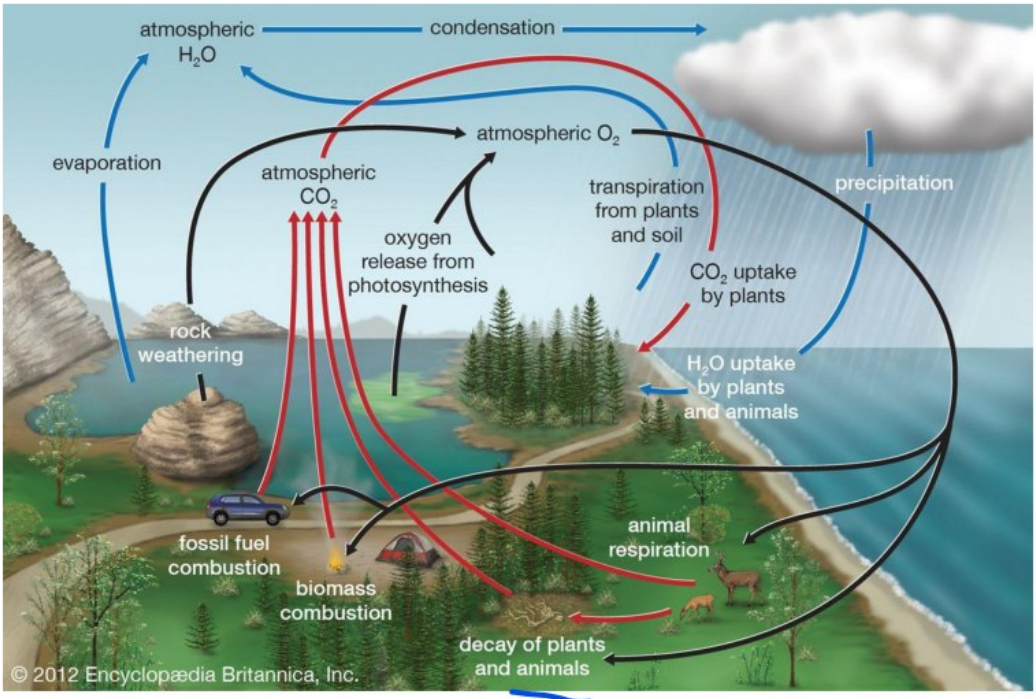


Fig: The Oxygen Cycle

Fig. 30 — Fig: The Oxygen Cycle. *(bank p. 10)*

Answer. Answer: The oxygen cycle describes the movement of oxygen through the atmosphere, biosphere, hydrosphere and lithosphere, closely linked with both the carbon and water cycles. Atmospheric O2 is released by photosynthesis in green plants (oxygen release from photosynthesis) and ● by rock weathering; CO2 is taken up by plants during photosynthesis. O2 is consumed through animal and plant respiration, and through fossil fuel/biomass combustion — both ● processes release CO2 back to the atmosphere. Water and oxygen exchange is also linked through evaporation, transpiration from plants and soil, ● condensation and precipitation — connecting the oxygen cycle with the hydrologic cycle. Decay of dead plants and animals also consumes oxygen (via decomposer respiration) and releases CO2.

QQ10. What do you mean by Tolerance Curve? Draw the Tolerance Curve and explain each part of the

2 marks · from the Unit 3 bank

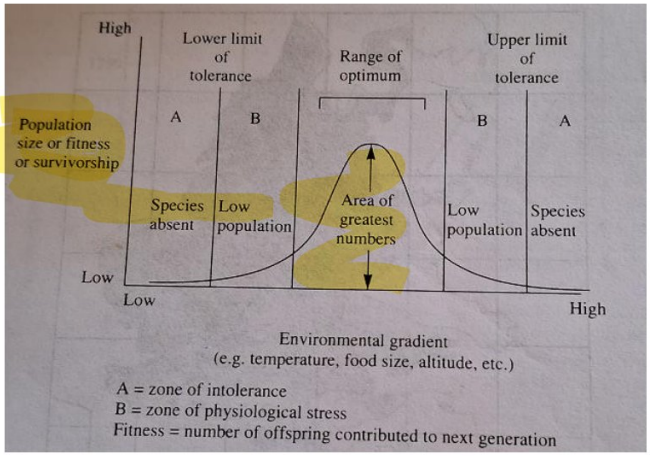


Fig. 31 — Fig: A schematic tolerance curve for a single species population existing on a single environmental gradient. *(bank p. 10)*

Answer. curve. [7] Answer: Tolerance refers to the range of physical conditions (e.g., temperature, moisture, pH, altitude) within which an organism can survive. If we plot some measure of success of a species population (such as population density or fitness) against an environmental gradient, a tolerance curve is generated — also known as the "universal tolerance curve." It allows predictions about where a species will be found. Range of Optimum: Within this range, the majority of individuals of the species population can survive ● and reproduce; a large population size is maintained since conditions along this part of the gradient are ideal for the species (the "area of greatest numbers"). Zone of Physiological Stress: Beyond the range of optimum, towards the high and low ends of the ● gradient, most individuals suffer increasing physiological stress. Conditions here are less than optimal — most individuals can stay alive but cannot function efficiently nor reproduce (marked as Zone B — low population). Zone of Intolerance: At the extremes of the gradient, no individuals of the species can survive because ● physical conditions or lack of food resources are too extreme to permit survival — the species is absent here (marked as Zone A). Axes: The x-axis represents the range of the environmental factor (low to high); the y-axis represents ● population size, fitness or survivorship (number of offspring contributed to the next generation).

QQ11. Describe the Hydrologic (Water) Cycle

7 marks · from the Unit 3 bank

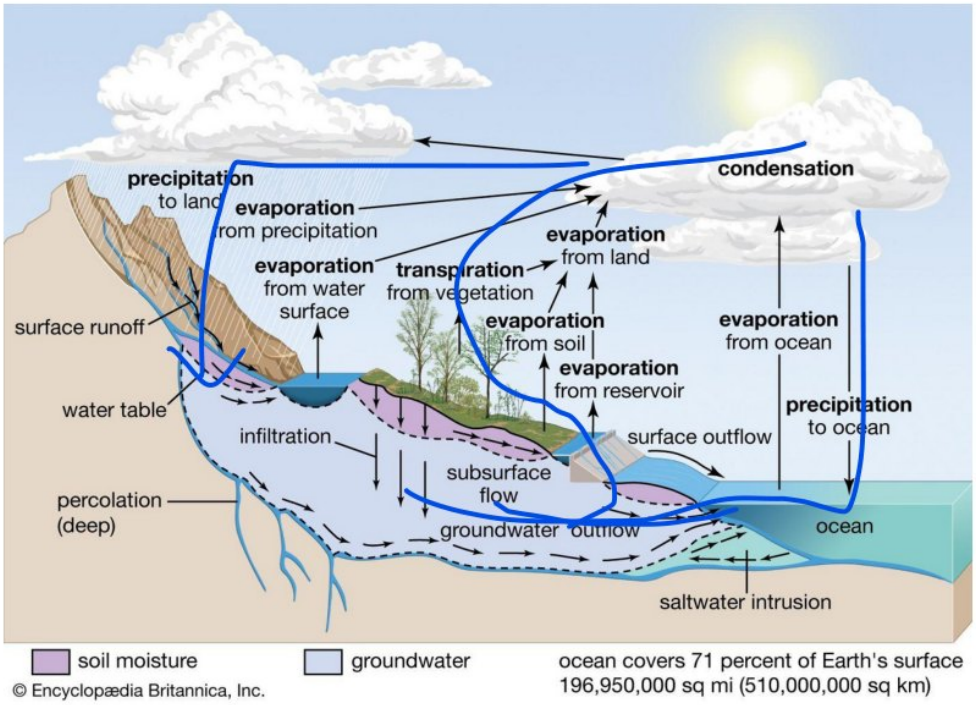


Fig. 32 — Fig: The Hydrologic (Water) Cycle. *(bank p. 11)*

Answer. Answer: The hydrologic cycle describes the continuous movement of water within the Earth and atmosphere, driven by solar energy. The ocean covers 71 percent of the Earth's surface (196,950,000 sq. mi / 510,000,000 sq. km) and is the largest reservoir of water. Evaporation & Transpiration: Water changes from liquid to vapour from oceans, water surfaces, land, ● soil and reservoirs; plants also release water vapour through transpiration from vegetation. Condensation: Water vapour rises and cools, forming clouds. ● Precipitation: Water falls back to Earth as rain/snow, either to land or directly to the ocean. ● Runoff & Infiltration: Water on land moves as surface runoff into rivers/streams, or infiltrates the soil ● (infiltration) to recharge the water table; water percolates deeper into the ground and moves as subsurface flow and groundwater outflow, eventually reaching the ocean (with saltwater intrusion possible near coastlines). — End of Question Bank —

UNIT 4 · Air and Noise Pollution

7 diagram-based questions · 9 diagrams in this unit

QQ1. Discuss the characteristics, primary source, and harmful effects of the criterion air pollutants

7 marks · from the Unit 4 (Air & Noise) bank

Ozone (O_3)	Key component of photochemical smog (tropospheric); criteria pollutant only in the troposphere	Formed via reaction of NO_2 and hydrocarbons in sunlight — common where traffic and sunshine are heavy	Respiratory irritation; damages crops and materials
Particulate matter (PM)	Airborne solid/liquid particles (soot, dust, smoke, fumes, mists), especially < 10 µm	Power plants, industrial processes, residential heating, vehicles	Inhaled deep into lungs; some (asbestos, soot) are carcinogenic
Lead (Pb)	Airborne particulates < 0.5 µm, very toxic	Oil refining, smelting, industrial activity (historically leaded gasoline)	Learning disabilities, seizures, even death, particularly in children

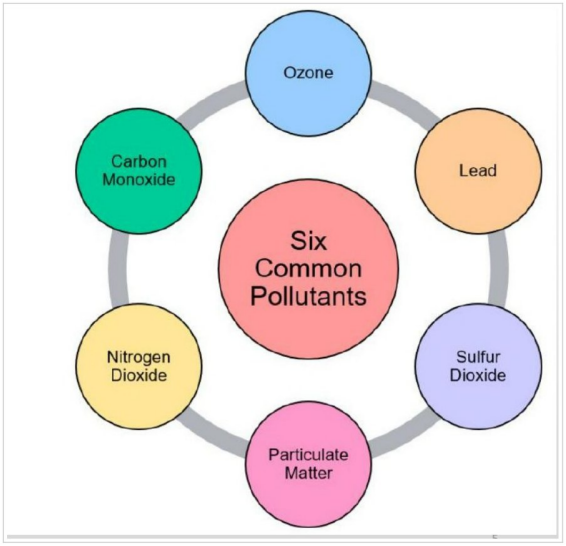


Fig. 33 — Fig. 1 — The six common (criteria) air pollutants designated by the EPA under the Clean Air Act / NAAQS. (bank p. 5)

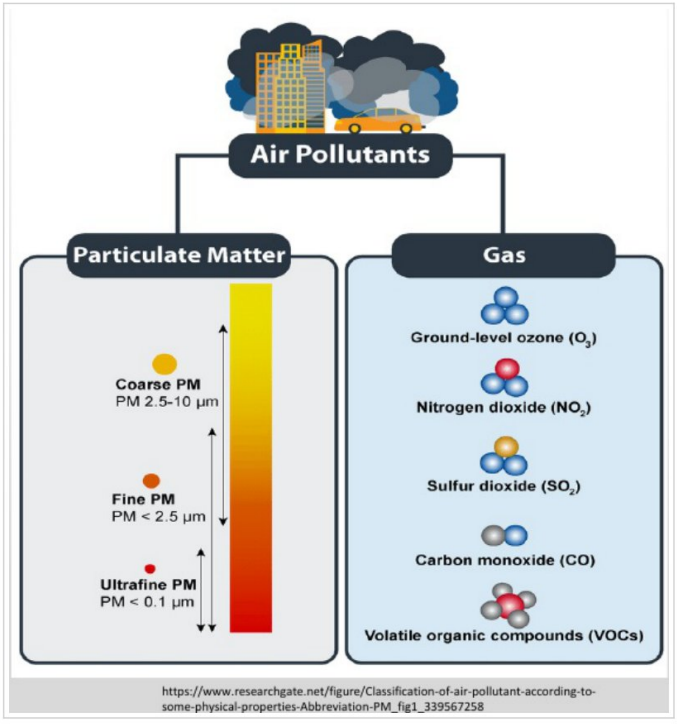


Fig. 34 — Fig. 2 — Classification of air pollutants into particulate matter (coarse/fine/ultrafine PM) and gaseous pollutants. (bank p. 6)

Answer. The U.S. EPA designates six "criteria" pollutants whose atmospheric concentrations serve as indicators of overall air quality: Pollutant Characteristics Primary Source Harmful Effects Carbon Odourless, invisible gas formed by Gasoline-powered highway vehicles Displaces oxygen in the bloodstream → monoxide incomplete combustion; most abundant (primary); residential heating and industrial asphyxiation at high (CO) criteria pollutant processes concentrations/exposure Sulfur Colourless gas with sharp, choking odour Combustion of coal/oil containing sulfur, Eye/throat irritation, lung damage; reacts dioxide mainly power-generating plants with O_2 and water vapour to form sulfuric (SO_2) acid → acid rain Nitrogen Pungent, irritating gas High-temperature combustion — coal power Pulmonary edema; forms nitric acid (acid dioxide plants and gasoline engines rain) and drives photochemical smog (NO_2) formation Ozone Key component of photochemical smog Formed via reaction of NO_2 and hydrocarbons Respiratory irritation; damages crops and (O_3) (tropospheric); criteria pollutant only in in sunlight — common where traffic and materials the troposphere sunshine are heavy Particulate Airborne solid/liquid particles (soot, dust, Power plants, industrial processes, residential Inhaled deep into lungs; some (asbestos, matter smoke, fumes, mists), especially < 10 heating, vehicles soot) are carcinogenic (PM) µm Lead (Pb) Airborne particulates < 0.5 µm, very Oil refining, smelting, industrial activity Learning disabilities, seizures, even death, toxic (historically leaded gasoline) particularly in children

QQ2. What is acid rain? Discuss with reactions, the harmful effects of acid rain

7 marks · from the Unit 4 (Air & Noise) bank

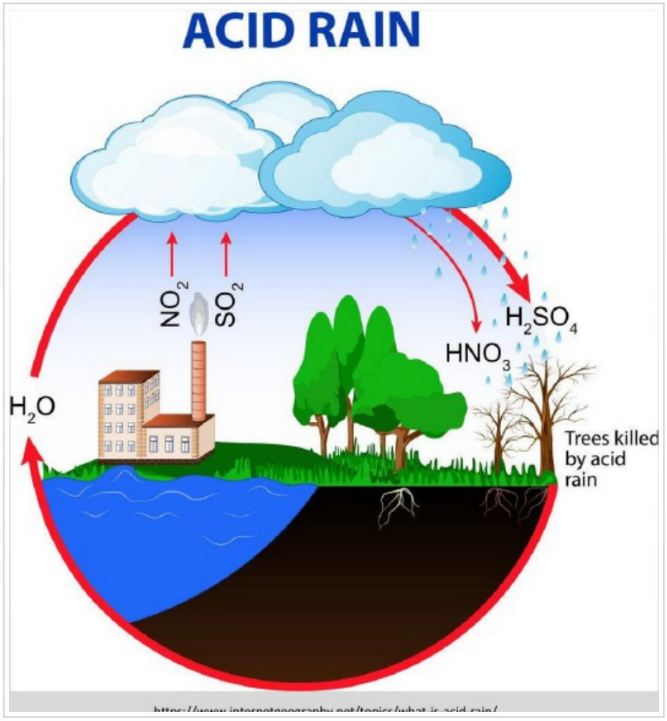


Fig. 35 — Fig. 3 — Formation cycle of acid rain: SO_2 and NO_2 react with atmospheric moisture to form H_2SO_4/HNO_3, which fall as acid rain and kill vegetation. *(bank p. 7)*

Answer. Acid rain is precipitation that has become abnormally acidic due to the reaction of atmospheric sulfur dioxide (SO_2) and nitrogen dioxide (NO_2) — emitted mainly from power plants and vehicles — with oxygen and water vapour in the atmosphere. Formation reactions: $SO_2 + O_2 + H_2O \rightarrow H_2SO_4$ (sulfuric acid) $NO_2 + H_2O \rightarrow HNO_3$ (nitric acid) These acids fall to the ground as "acid rain," which can also deposit as dry acidic particles. Harmful effects: Harms or destroys fish and plant life in lakes and forests. Corrodes metals and deteriorates exposed surfaces of buildings and monuments (e.g., the Taj Mahal). Kills trees and degrades soil quality. Contributes to respiratory problems via associated SO_2/NO_2 exposure.

QQ6. Discuss the operational principle of electrostatic precipitator. Comment on its efficiency

7 marks · from the Unit 4 (Air & Noise) bank

- An electrostatic precipitator can remove particulates as small as 1 µm with an efficiency exceeding 9
- Its effectiveness in removing fly ash from combustion gases of fossil-fuel furnaces accounts for its wic stations.

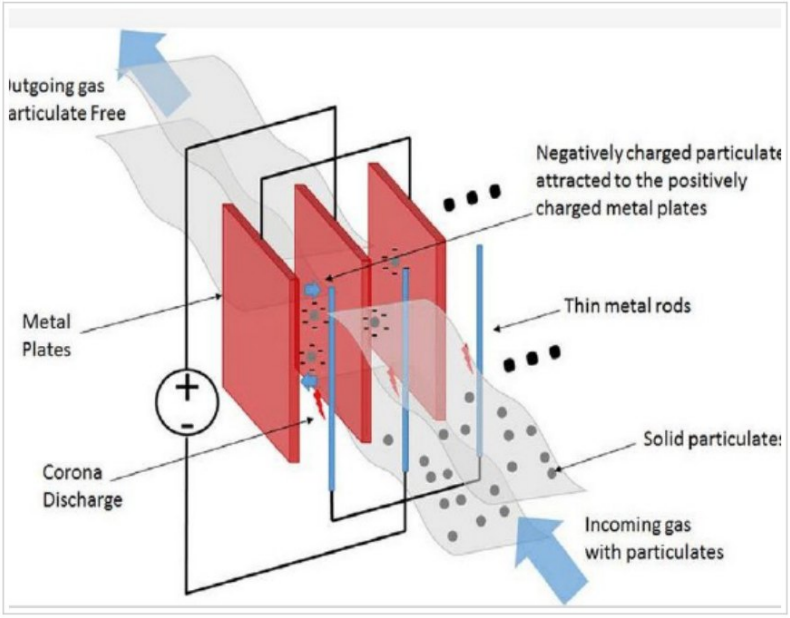


Fig. 36 — Fig. 4 — Working of an electrostatic precipitator: charged metal plates attract negatively charged particulates via corona discharge, leaving clean *(bank p. 8)*

Answer. Operating principle: 1. Particles suspended in the polluted airstream are given an electric charge as they enter the unit. 2. A high DC voltage (as much as 100,000 volts) is applied to discharge electrode wires, which hang between rows of large, grounded (positively charged) collection plates arranged vertically and in parallel inside a boxlike structure. 3. The negatively charged wires create a corona discharge that charges the dust particles, which are then attracted to and trapped on the oppositely (positively) charged collection plates. 4. Collected particles form a dust layer (~6 mm thick) on the plates, which is periodically dislodged by mechanically shaking or "rapping" the plates. 5. Dislodged particles fall into a hopper at the bottom of the unit and are removed for disposal. Efficiency: An electrostatic precipitator can remove particulates as small as 1 µm with an efficiency exceeding 99 percent. Its effectiveness in removing fly ash from combustion gases of fossil-fuel furnaces accounts for its widespread use at power stations. outgoing gas.

QQ7. Write a short note on flue gas desulfurization

7 marks · from the Unit 4 (Air & Noise) bank

- **Wet FGD:** flue gases contact a liquid or slurry absorbent; SO_2 dissolves in or reacts with the absorbent and becomes trapped.
- **Dry FGD:** the absorbent is dry pulverized lime or limestone; the resulting solid particles are removed using baghouse filters. Dry systems offer cost/energy savings and easier operation but need higher chemical consumption and work only with low-sulfur coal flue gases.

Key chemical reactions:

```
Limestone scrubbing: CaCO₃ + SO₂ → CaSO₃ + CO₂
Lime scrubbing: Ca(OH)₂ + SO₂ → CaSO₃ + H₂O
Oxidation: 2 CaSO₃ + O₂ → 2 CaSO₄ (gypsum)
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- Neither calcium sulfite nor calcium sulfate is very soluble in water; both precipitate out as a slurry by gravity settling.
- The resulting thick slurry, called FGD sludge, creates a significant disposal problem, and millions of tons are generated each year — though FGD does reduce ambient SO_2 and mitigate acid rain.

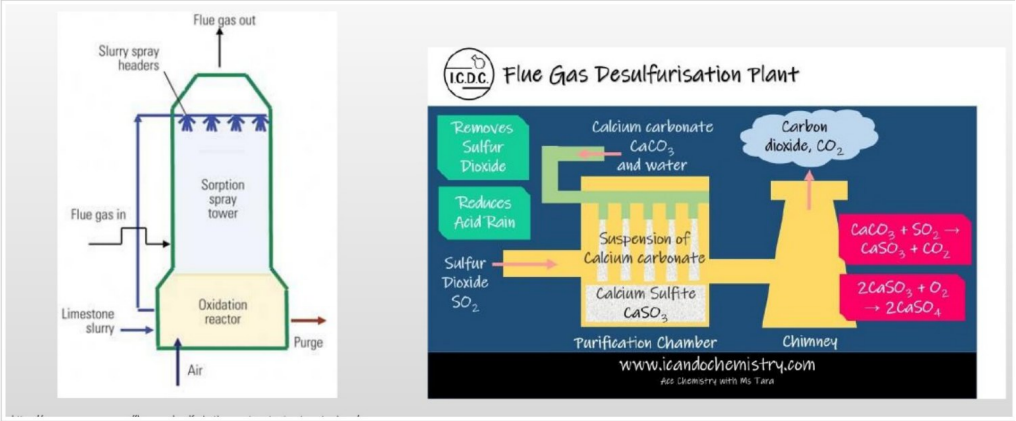


Fig. 37 — Fig. 5 — Flue gas desulfurization: (left) wet scrubbing sorption spray tower; (right) limestone-based FGD plant removing SO_2 and reducing acid rain. *(bank p. 9)*

Answer. Flue gas desulfurization (FGD) is an absorption process used to control sulfur dioxide in flue gas from fossil-fuel power plants. FGD systems may involve wet or dry scrubbing. Wet FGD: flue gases contact a liquid or slurry absorbent; SO_2 dissolves in or reacts with the absorbent and becomes trapped. Dry FGD: the absorbent is dry pulverized lime or limestone; the resulting solid particles are removed using baghouse filters. Dry systems offer cost/energy savings and easier operation but need higher chemical consumption and work only with low-sulfur coal flue gases. Key chemical reactions: Limestone scrubbing: $CaCO_3 + SO_2 \rightarrow CaSO_3 + CO_2$ Lime scrubbing: $Ca(OH)_2 + SO_2 \rightarrow CaSO_3 + H_2O$ Oxidation: $2\ CaSO_3 + O_2 \rightarrow 2\ CaSO_4$ (gypsum) Neither calcium sulfite nor calcium sulfate is very soluble in water; both precipitate out as a slurry by gravity settling. The resulting thick slurry, called FGD sludge, creates a significant disposal problem, and millions of tons are generated each year — though FGD does reduce ambient SO_2 and mitigate acid rain.

QQ8. Write a short note on noise pollution

7 marks · from the Unit 4 (Air & Noise) bank

- Avoid playing speakers, radios and television loudly.
- Avoid unnecessary bursting of crackers.
- Avoid unnecessary honking; service vehicles regularly to prevent noisy engines.
- Plant trees around buildings and along roads to buffer noise.
- Separate noise sources (industries, airports) from residential areas.
- Enforce fines for exceeding permissible noise limits and monitor ambient noise levels.



Fig. 38 — Fig. 6 — Typical loudness (dB) of common sounds, from breathing (0 dB) to a rocket launch (140+ dB). (bank p. 10)

Answer. Definition: Noise pollution is unwanted or excessive sound that has deleterious effects on human health, wildlife, and environmental quality. Types: Man-made noise: generated by human activities — construction work, automobiles, household appliances (30 dB–140 dB). Environmental noise: from natural sources — animal sounds, thunderstorms, cyclones, hurricanes (up to 140 dB). Sources: bursting of crackers, loud radios/speakers/TV, construction activities, industrial machinery, automobile horns and engines, aircraft (up to 130 dB), home appliances (mixers, washing machines, pressure cookers), animal sounds, and severe weather. Measurement: Sound waves are vibrations of air molecules; sound cannot travel in a vacuum. Loudness (sound pressure level, SPL) is measured in decibels (dB) using $L_p = 20 \log_{10}(p_{rms}/p_{ref})$, where $p_{ref} = 20$ µPa. Humans can detect 0–140 dB; 120–140 dB causes pain. Effects: hearing loss, headaches, high blood pressure, heart attacks, hypertension, depression, anxiety, fatigue and stress, sleep disorders, loss of concentration, and disturbance/harm to wildlife. Prevention/Reduction: Avoid playing speakers, radios and television loudly. Avoid unnecessary bursting of crackers. Avoid unnecessary honking; service vehicles regularly to prevent noisy engines. Plant trees around buildings and along roads to buffer noise. Separate noise sources (industries, airports) from residential areas. Enforce fines for exceeding permissible noise limits and monitor ambient noise levels.

QQ9. Describe the main techniques used to control particulate and gaseous air pollutants at industrial facilities

7 marks · from the Unit 4 (Air & Noise) bank



Fig. 39 — Fig. 7 — A cyclone separator: the swirling gas stream forces coarse particles to the outer wall and down into a collection hopper. *(bank p. 11)*



ling gas stream forces coarse particles to the outer w



Fig. 40 — Fig. 8 — A baghouse: dirty air is blown through fabric filter bags; particulates are trapped while clean air exits at the top. *(bank p. 11)*

Answer. (a) Particulate control: Common equipment includes cyclones, scrubbers, electrostatic precipitators, and baghouse filters (see Cyclone: Removes particulates by spinning the dirty airstream in a spiral path inside a cylindrical chamber; larger particles are flung outward by inertia, slide down the wall into a dust hopper, while cleaned air exits through an inner cylinder.

Cyclones are best at coarse particles (> 20 µm, ~90% efficiency) and are typically used as pre-cleaners before ESPs/baghouses. Scrubber (wet): Traps suspended particles by direct contact with a spray of water or liquid; particulates collide with and are entrained by fine droplets. Spray-tower scrubbers remove ~90% of particles > 8 µm. Electrostatic precipitator (ESP): Charges particles electrically and collects them on oppositely charged plates (see Q6, Section B). Baghouse filter: Fabric-bag filters trap particulates as air is blown through; nearly 100% efficient for particles as small as 1 µm. (b) Gaseous pollutant control: Gaseous criteria pollutants, VOCs, and other air toxics are controlled by three basic techniques — used singly or combined — plus a fourth developing technique for CO_2: 1. Absorption: Transfers a gaseous pollutant from air into a contacting liquid (e.g., water), which serves as a solvent or reacts chemically with the pollutant. Wet/packed scrubbers achieve 90–95% gas-removal efficiency; used at fertilizer plants (ammonia removal), glass plants (hydrogen fluoride), chemical plants (water-soluble solvents), and rendering plants (odour control). Flue gas desulfurization (FGD) is a major example (see Q7). 2. Adsorption: A surface phenomenon where gas molecules are attracted to and held on a solid surface (sorbed). Activated carbon (heated charcoal) is the most common adsorbent, due to its high porosity and surface-area-to-volume ratio; used for odour control, VOC removal, and solvent recovery, achieving > 95% efficiency. 3. Incineration (combustion): Converts VOCs and hydrocarbon pollutants to CO_2 and water via rapid oxidation in a special incinerator called an afterburner, which must provide sufficient turbulence, burning time, and temperature. Catalysts (e.g., platinum) allow complete oxidation at lower temperatures. Used at petroleum refineries, paint-drying facilities, and paper mills. 4. Carbon sequestration: Long-term storage of CO_2 underground or in natural sinks (see Topic 10 below).

QQ10. Explain carbon sequestration and how carbon credits help monetise it, as a policy tool for reducing air

2 marks · from the Unit 4 (Air & Noise) bank

- **Geosequestration (carbon capture and storage):** CO_2 is separated from power-plant flue gases (a costly pro pumped into underground geologic "reservoir" layers such as depleted oil/gas fields or deep saline aquifers, wher remain stored for potentially thousands of years.
- **Biological sequestration:** Relies on natural carbon sinks — forests, oceans, grasslands, and wetlands — that at from the atmosphere; deforestation and ocean pollution diminish this natural capacity.
- **Carbon credits:** A certificate representing removal of one metric tonne of CO_2 from the atmosphere. Companies governments purchase carbon credits to meet climate commitments (becoming "carbon neutral").
- As a policy measure, initiatives like India's Energy Conservation (Amendment) Bill, 2022 aim to develop a domest market, indirectly benefiting farmers through soil-carbon-linked incentives, since higher soil carbon also improves water retention, and reduces flood/drought cycles.

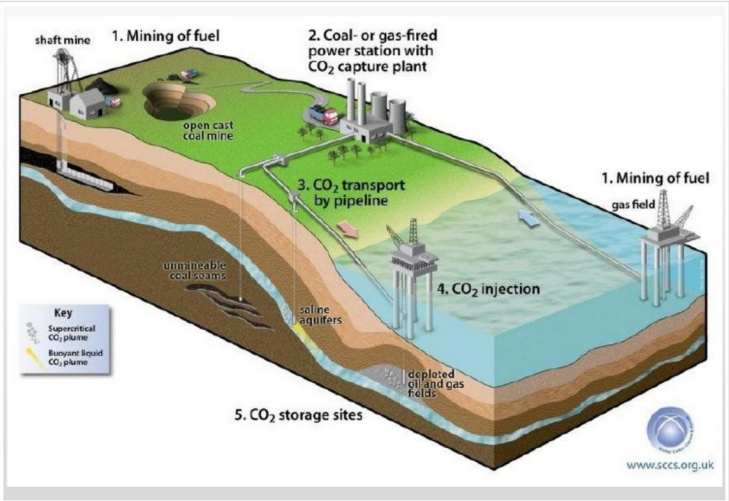


Fig. 41 — Fig. 9 — Geosequestration: CO_2 captured from a power station is transported by pipeline and injected into underground storage sites (saline aquifers, (bank p. 12)

Answer. pollution. (7) The best way to reduce atmospheric CO_2 is to use energy more efficiently and shift to alternative energy sources (nuclear, wind, tidal, solar); carbon sequestration serves as an additional tool. Geosequestration (carbon capture and storage): CO_2 is separated from power-plant flue gases (a costly process) and pumped into underground

geologic "reservoir" layers such as depleted oil/gas fields or deep saline aquifers, where it can remain stored for potentially thousands of years. Biological sequestration: Relies on natural carbon sinks — forests, oceans, grasslands, and wetlands — that absorb CO_2 from the atmosphere; deforestation and ocean pollution diminish this natural capacity. Carbon credits: A certificate representing removal of one metric tonne of CO_2 from the atmosphere. Companies and governments purchase carbon credits to meet climate commitments (becoming "carbon neutral"). As a policy measure, initiatives like India's Energy Conservation (Amendment) Bill, 2022 aim to develop a domestic carbon market, indirectly benefiting farmers through soil-carbon-linked incentives, since higher soil carbon also improves soil health, water retention, and reduces flood/drought cycles. depleted oil/gas fields).

UNIT 5 · Water Pollution Abatement and Solid Waste Management

9 diagram-based questions · 11 diagrams in this unit

QA5. The results from a BOD test, diluted by 100, are given in the table. Calculate BOD5

2 marks · from the Unit 5 bank

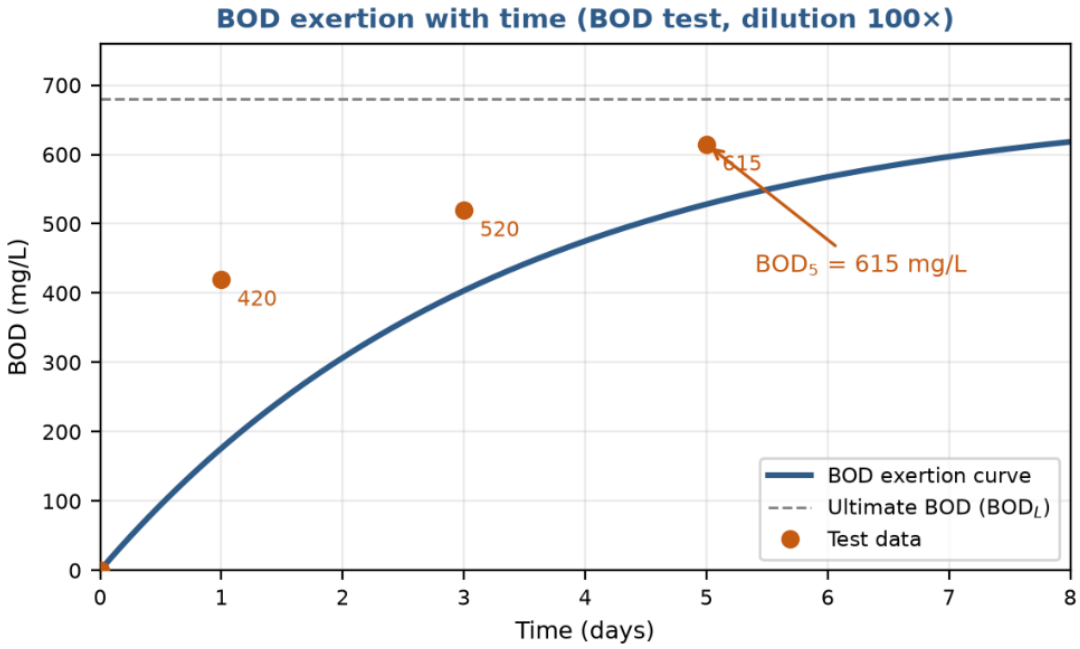


Fig. 42 — Fig. A5 - BOD exertion with time; BOD5 is read at day 5. (bank p. 3)

Answer. BOD5 = p (DOi - DOf), where p = dilution factor (100) and DOi, DOf are the initial and final DO concentrations. Time (days) Diluted sample DO (mg/L) BOD (calculated) (mg/L) 0 7.95 - 1 3.75 420 3 2.75 520 5 1.80 615 Therefore BOD5 = 100 x (7.95 - 1.80) = 615 mg/L. Biology and Environmental Science

QA6. What is eutrophication?

2 marks · from the Unit 5 bank

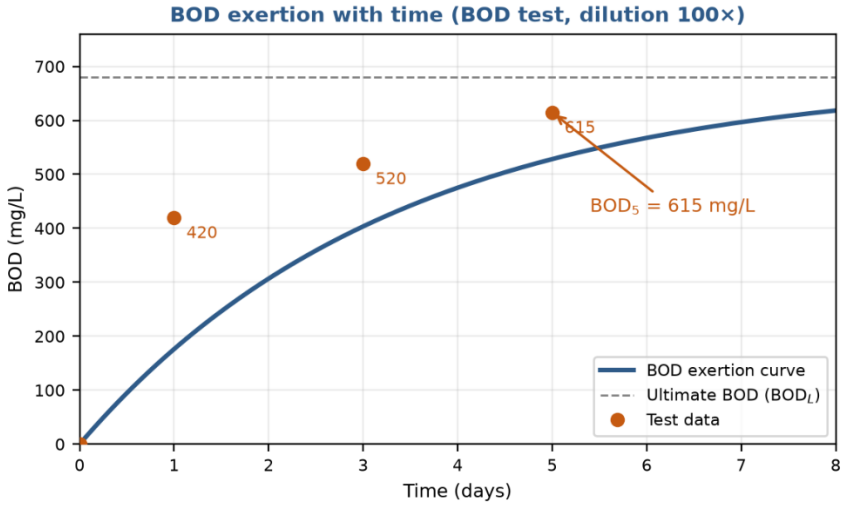


Fig. A5 - BOD exertion with time; BOD$_5$ is read at day 5.

A6. What is eutrophication?

Eutrophication is the process by which a lake changes from a clean, clear condition - with a relatively low concentration of dissolved nutrients and a balanced aquatic community - to a nutrient-rich, algae-filled state and thence to an oxygen-deficient, waste-filled condition. Excess nitrates and phosphates from

Fig. 43 — Fig. A6 - Stages of eutrophication of a water body. (bank p. 3)

Answer. Eutrophication is the process by which a lake changes from a clean, clear condition - with a relatively low concentration of dissolved nutrients and a balanced aquatic community - to a nutrient-rich, algae-filled state and thence to an oxygen-deficient, waste-filled condition. Excess nitrates and phosphates from sewage and fertiliser run-off promote algal blooms; when the algae die, microbial decomposition consumes dissolved oxygen, and anaerobic organisms then release gases such as methane and hydrogen sulphide. Eutrophication is a naturally occurring, slow and inevitable process, but when accelerated by human activity it is called cultural eutrophication.

QA9. Mention the goals of circular economy

2 marks · from the Unit 5 bank



Fig. 44 — Fig. A9 - Linear economy versus circular economy. (bank p. 4)

Answer. The goal of the circular economy (CE) is to optimize circle-system functions by reducing the resources exiting the system - that is, to minimise waste and to treat goods, materials and waste as secondary raw materials that can be recycled and reused. This sets it

apart from the linear 'take-make-use-dispose' economy. Moving towards a greater circular economy helps meet the resource-efficiency goals of the strategy for smart, inclusive and sustainable growth; in wastewater treatment it promotes reuse of treated effluent for irrigation, industry, toilet flushing and groundwater replenishment. Biology and Environmental Science

QB1. What are the three phases of waste water treatment? Discuss, with diagrams, the

7 marks · from the Unit 5 bank

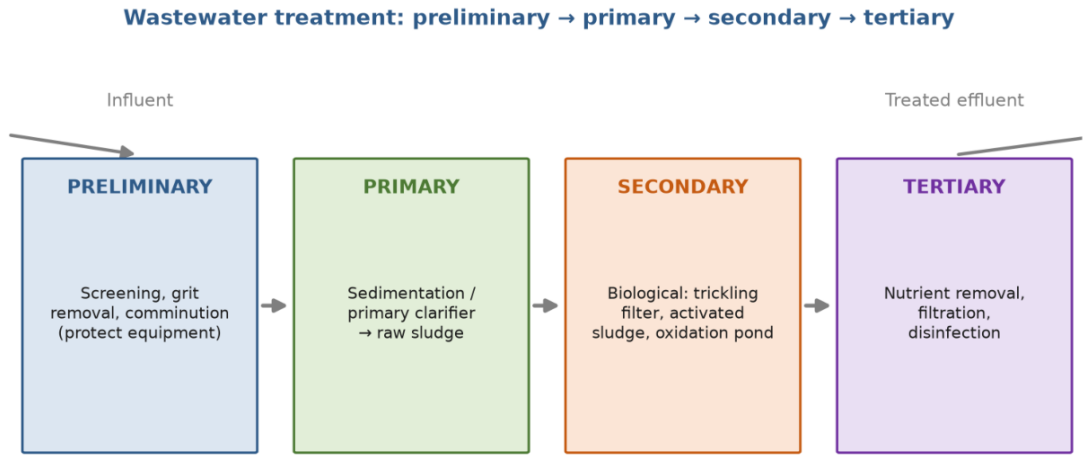


Fig. 45 — Fig. B1a - The phases of wastewater treatment. (bank p. 5)

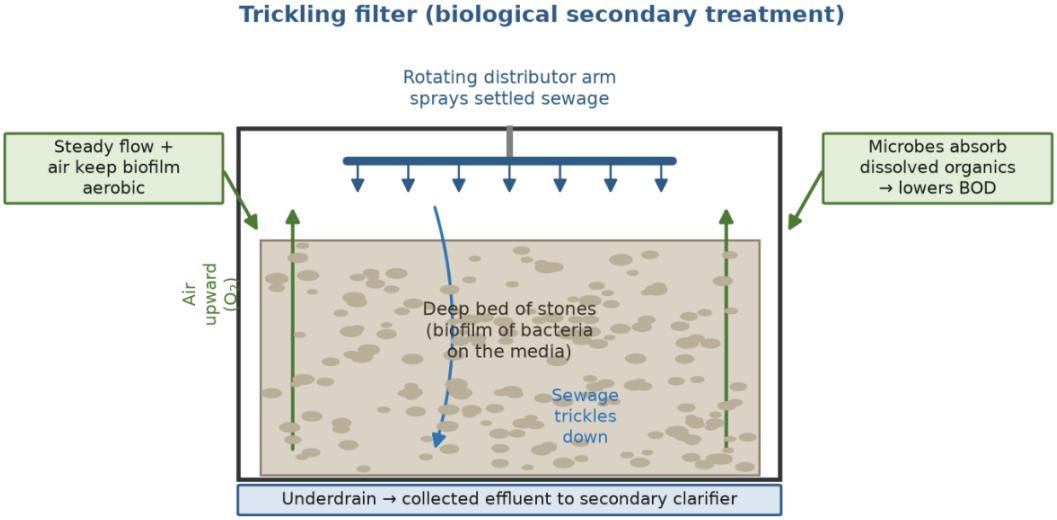


Fig. 46 — Fig. B1b - Cross-section of a trickling filter. (bank p. 6)

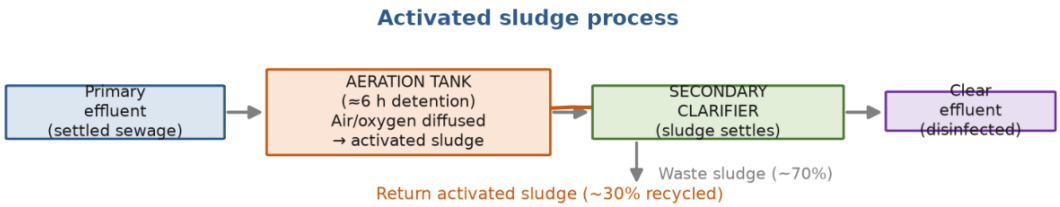


Fig. 47 — Fig. B1c - The activated sludge process (a second secondary-treatment method). (bank p. 6)

Answer. working principle of the trickling filter. Wastewater is treated in a sequence of phases, each removing a different class of impurity. Before the main phases, preliminary treatment (screening, grit removal, comminution) protects the plant's equipment and infrastructure. G Primary treatment removes material that will either float or readily settle out by gravity - screening, comminution, grit removal and sedimentation in primary clarifiers (about two hours of detention time). The settled solids form raw or primary sludge; grease and floating matter are skimmed off mechanically. G Secondary treatment removes the soluble organic matter that escapes primary treatment, using biological processes in which microbes consume the organic impurities as food and convert them to carbon dioxide, water and energy, thereby lowering the BOD. The three basic methods are the trickling filter, the activated sludge process and the oxidation pond; a fourth, less common method is the rotating biological contactor. G Tertiary (advanced) treatment removes residual suspended solids, nutrients (nitrogen and phosphorus), toxic compounds and dissolved inorganics - for example effluent polishing by granular-media filters, nutrient removal, carbon adsorption and disinfection. Trickling filter - working principle. A trickling filter is simply a tank filled with a deep bed of stones. Settled sewage is sprayed continuously over the top of the stones and trickles to the bottom, where it is collected for further treatment. As the wastewater trickles down, bacteria gather and multiply on the stones, forming a biofilm. The steady flow of sewage over these growths allows the microbes to absorb the dissolved organics, thus lowering the BOD of the sewage. Air circulating upward through the spaces among the stones provides sufficient oxygen for the metabolic processes. Settling tanks called secondary clarifiers follow the filter and remove microbes washed off the stones; two or more filters may be connected in series, and sewage may be recirculated to increase treatment efficiency. Biology and Environmental Science For comparison, in the activated sludge process settled sewage is mixed with recycled sludge in an aerated tank (about six hours' detention), where diffused air or mechanical mixers supply oxygen; the mixture then flows to a secondary clarifier, about 30 per cent of the settled sludge being returned to the aeration tank and the rest wasted.

QB2. What are biological oxidation ponds? Discuss their working principle with a proper

7 marks · from the Unit 5 bank

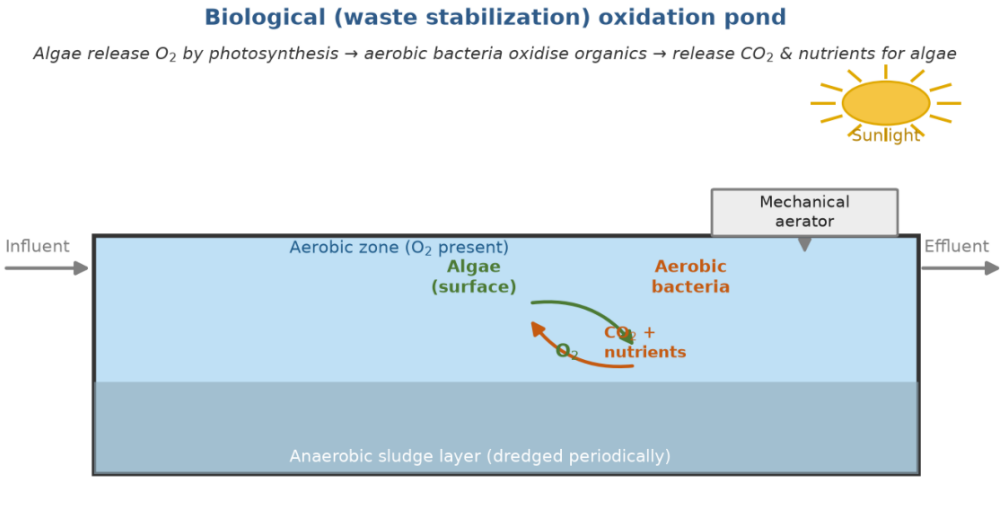


Fig. 48 — Fig. B2 - Cross-section of a biological oxidation (stabilization) pond. *(bank p. 7)*

Answer. diagram. Oxidation ponds, also called lagoons or stabilization ponds, are large shallow ponds designed to treat wastewater through the interaction of sunlight, bacteria and algae. They are a biological secondary-treatment method. Working principle. Algae grow using energy from the sun, carbon dioxide and the inorganic compounds released by bacteria in the water. During photosynthesis the algae release the oxygen needed by aerobic bacteria; the bacteria in

turn oxidize the dissolved organic matter, releasing carbon dioxide and nutrients that the algae use. This symbiotic algae-bacteria cycle both stabilizes the organic waste and keeps the pond aerobic. Mechanical aerators are sometimes installed to supply additional oxygen, which reduces the required size of the pond. Sludge deposits in the pond must eventually be removed by dredging or scooping, and algae remaining in the effluent can be removed by filtration or by Biology and Environmental Science a combination of chemical treatment and settling.

QB3. What is Integrated Solid Waste Management (ISWM)? Explain elaborately the four

7 marks · from the Unit 5 bank

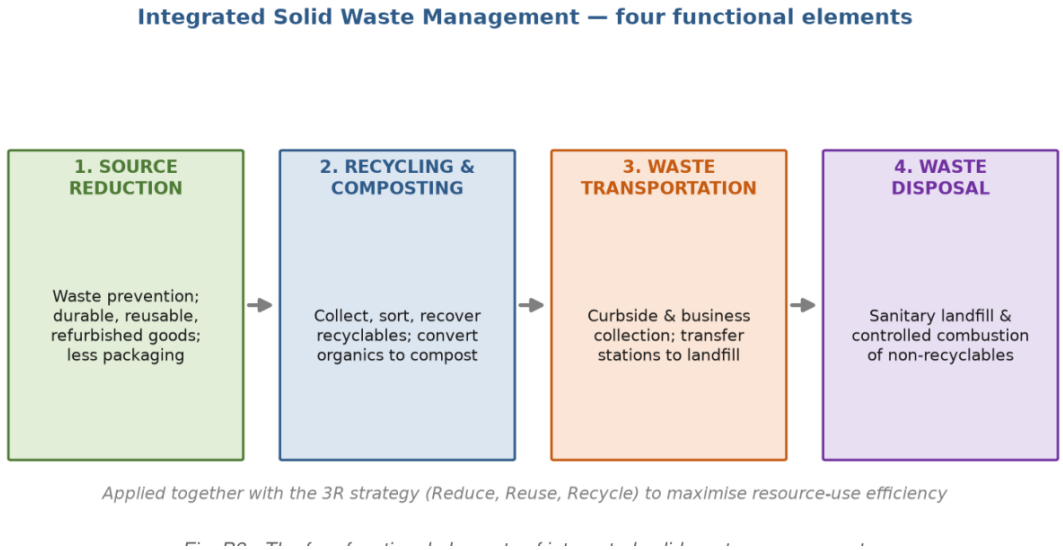


Fig. 49 — Fig. B3 - The four functional elements of integrated solid waste management. (bank p. 8)

Answer. functional elements of ISWM. Integrated solid waste management (ISWM) is the strategic approach to the sustainable management of solid wastes, covering all sources and all aspects - waste generation, segregation, transfer, sorting, treatment, recovery and disposal - in an integrated manner, with emphasis on maximising resource-use efficiency. It includes the collective management of all types of waste and implementation of the 3R (Reduce, Reuse, Recycle) policies. An efficient ISWM system evaluates local conditions and needs, then chooses, mixes and applies the most suitable activities. The four functional elements are: 1 Source reduction (waste prevention) - reduces unnecessary waste generation through products designed for recycling, durable and sustainable goods, reusable products and packaging, refurbishing goods to prolong product life, redesign to use less packaging, and reducing food spoilage. It lowers waste handling, transport and disposal costs and reduces methane generation. 2 Recycling and composting - recycling includes the accumulation, sorting and recovery of recyclables and their reprocessing into new products; composting converts organic waste into soil additives. Both create jobs, divert material from the waste stream and significantly reduce greenhouse-gas emissions. 3 Waste transportation - collection of waste from curbside and businesses, and from transfer stations where waste is concentrated and reloaded onto other vehicles for delivery to the landfill; it must be integrated systematically with the other activities. 4 Waste disposal - activities for waste that is not recycled, in particular sanitary landfilling and combustion. Landfills must be properly designed, well-constructed and systematically managed. Biology and Environmental Science Importance. With rapid population growth and economic development, waste generation keeps rising, putting pressure on society's ability to process and dispose of it. Inappropriately managed solid waste pollutes water, attracts rodents and insects, blocks drains (increasing floods) and raises greenhouse-gas emissions. A comprehensive ISWM system for efficient collection, transportation and systematic disposal - together with waste reduction and increased recycling - significantly reduces all these

problems.

QB4. Describe the major components of a hazardous waste management plan

7 marks · from the Unit 5 bank

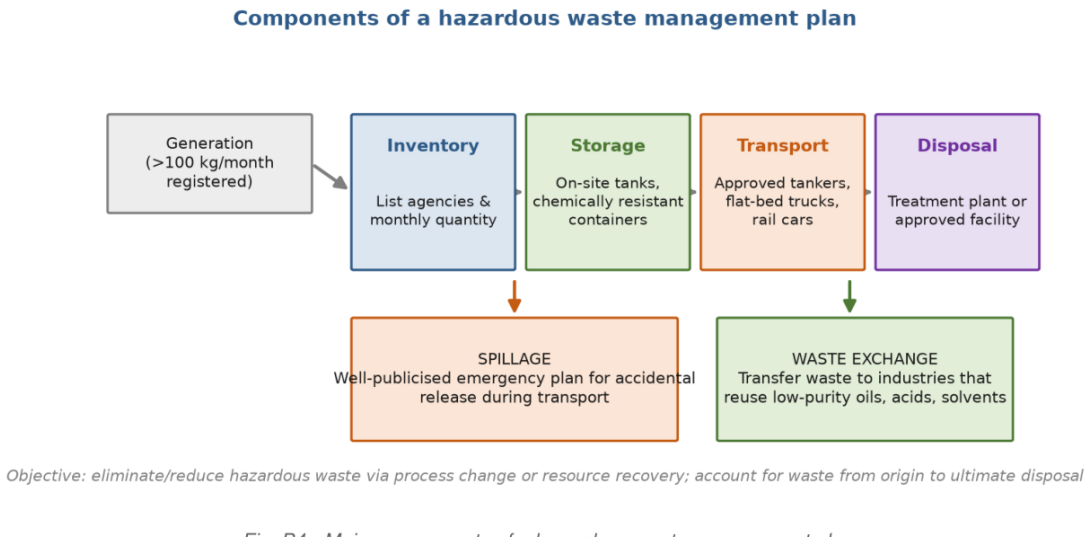


Fig. 50 — Fig. B4 - Major components of a hazardous waste management plan. *(bank p. 9)*

Answer. Hazardous waste is defined (USEPA) as any solid, liquid or gaseous substance which, by reason of any physical, chemical, reactive, toxic or infectious characteristics, causes danger to health or the environment when improperly treated, stored, transported or disposed of. Sources include industries, petroleum refineries, coke ovens, blast-furnace and scrubbing sludges, biological-waste laboratories, and thermal and nuclear power plants. The main objective of a hazardous-waste management plan is to eliminate or reduce hazardous waste through process changes or resource recovery, and to account for the waste from its origin to ultimate disposal. Its major components are: G Inventory - all industries producing more than 100 kg of hazardous waste per month must be registered, and an inventory of such generating agencies and the quantity produced per month prepared. G Storage - generating plants must equip themselves with special on-site tanks, chemically resistant containers or basins for storage of waste. G Transport - wastes must be transported by suitable tanker trucks, flat-bed trucks or rail cars to approved disposal sites. G Spillage - a well-publicised emergency plan to deal with unexpected spillage or accidental release during transport, to prevent environmental damage and public-health hazard. G Disposal - wastes transported to physical and chemical treatment plants for processing or concentration, or hauled directly to an approved hazardous-waste treatment facility for final disposal. G Waste exchange - when on-site recycling or recovery is not feasible, transfer waste from large industries to smaller ones able to reuse low-purity oils, acids, alkalies or solvents, or recover valuable metals and materials. Biology and Environmental Science

QB5. What is e-waste? Cite their common sources and composition. Discuss the steps

7 marks · from the Unit 5 bank

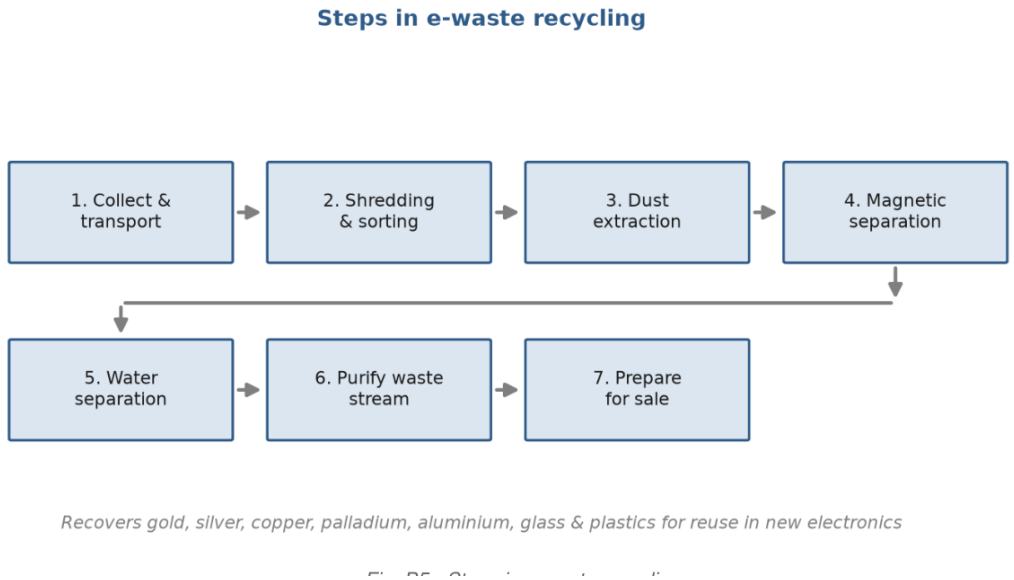


Fig. 51 — Fig. B5 - Steps in e-waste recycling. *(bank p. 10)*

Answer. involved in e-waste recycling. All items of electrical and electronic equipment and their parts that have been discarded by the user as waste, without the purpose of reuse or recycle, are called electronic waste (e-waste); stored used electronics are also considered e-waste. Improper processing of e-waste can lead to dangerous human-health effects and environmental pollution. Sources. Large household appliances (refrigerators/freezers, washing machines, dishwashers, televisions); small household appliances (toasters, coffee makers, irons, hairdryers); IT and telecommunications equipment (personal computers, telephones, mobile phones, laptops, printers, scanners, photocopiers); lighting equipment (fluorescent lamps); electrical/electronic tools (handheld drills, saws, screwdrivers); toys and sports equipment; monitoring and control instruments; and automatic dispensers. Composition. Valuable metals (gold, platinum, silver, palladium); useful metals (copper, aluminium, iron); hazardous substances (radioactive isotopes, mercury); toxic substances (dioxins); plastics (HIPS, ABS, PC, PPO); and glass (CRT glass of SiO2, CaO, Na). A single mobile phone contains more than 40 elements. Steps involved in e-waste recycling. 1 Collecting and transporting - recyclers place take-back booths or collection bins; when these fill, the e-waste is transported to recycling facilities. 2 Shredding and sorting - e-waste is broken into smaller pieces, then sorted and manually dismantled to retrieve different parts and separate core materials from components. 3 Dust extraction - the shredded particles are spread on a conveyor and broken down further; the dust is extracted and disposed of in an environmentally compliant manner. 4 Magnetic separation - a strong overhead magnet separates steel and iron from the other wastes. 5 Water separation - separates glass from plastic; lead-containing glass goes to smelters for batteries, X-ray tubes and new CRTs. 6 Purification of the waste stream - leftover metals are extracted from the plastics. 7 Preparing recycled materials for sale - the separated materials are prepared as raw materials for new electronics. Biology and Environmental Science

QB7. What is Electromagnetic Radiation (EMR)? Mention its two types. Write the impacts of

7 marks · from the Unit 5 bank

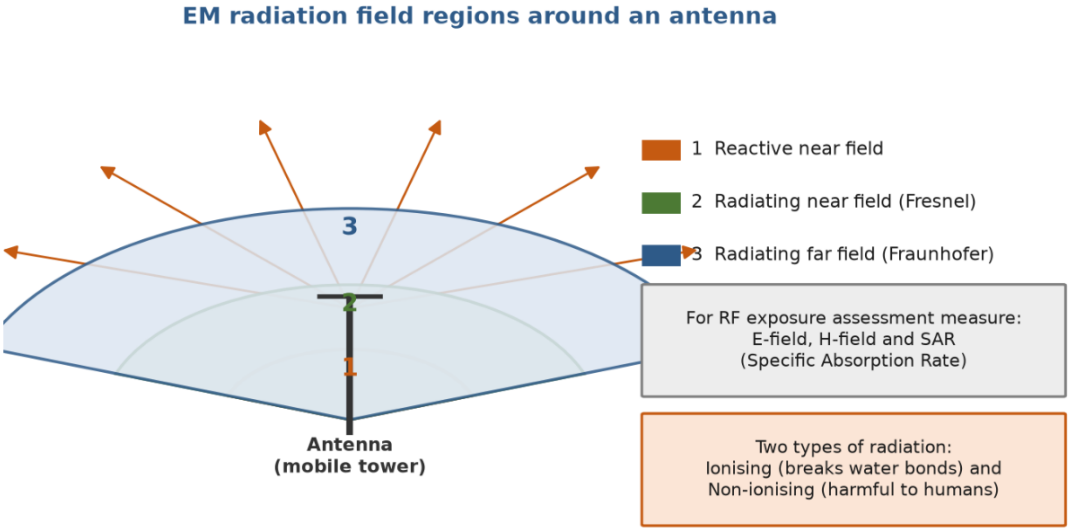


Fig. 52 — Fig. B7 - EM radiation field regions around an antenna. (bank p. 11)

Answer. EMR exposure on human health. Explain in brief the mitigation techniques. Electromagnetic radiation (EMR) is a fundamental form of energy that surrounds us everywhere - from the microwaves that heat our food to the radio waves that bring us music and news. With the rapid growth of mobile and wireless devices, telecom operators deploy many cellular towers that employ multiple antennas and transmit electromagnetic waves, so exposure to many EMF frequencies is increasing significantly. The EM field around an antenna is divided into regions - reactive near field, radiating near field (Fresnel) and radiating far field (Fraunhofer) - and RF-exposure assessment requires measuring the E-field, H-field and SAR (Specific Absorption Rate). Two types of radiation: G Ionising radiation - carries enough energy to break the chemical bonds of water and biological molecules. G Non-ionising radiation - does not break these bonds, but is still harmful to human beings. Biology and Environmental Science Impacts on human health. A high level of RF exposure heats body parts (tissues), because humans do not have strong heat-dissipating capability; parts such as the eyes and kneecaps are affected more. More severe effects can include behavioural abnormalities, cancer, physiological malfunctions (reproductive impairment, kidney failure), physical deformations and birth defects. Mitigation techniques. G Reduce transmit power - power density depends on the square of E/H, so lowering transmit power lowers the RF radiation level (though this reduces the coverage area). G Increase antenna height - the RF field strength at the observer point falls as the distance increases. G Higher antenna gain with lower transmit power can be used in close proximity to the antenna. G Basic health precautions - minimise use of mobile and wireless devices, do not continuously wear wireless (Bluetooth/Zigbee) earphones, keep children's use low, and use devices in safer zones far from base stations or cellular towers. G Field management - vertical positioning of current-carrying conductors on power-line structures produces lower magnetic fields; the electric-field component is easily shielded by most structures, though magnetic fields are difficult to contain. The ICNIRP has developed international guidelines and limits for EM radiation, and EM pollution can be tested with an ELF/EMF meter (a reading generally below 1.0 mG away from appliances is acceptable).

Part 4 · Diagrams worth being able to draw from memory

The paper rewards a labelled sketch wherever one is possible; these are the ones the banks themselves illustrate and the briefing names as high-yield. Practise each on a half page, labels only, no shading.

Unit	Diagrams to be able to reproduce
Unit 1	Lipid bilayer (A17) · enzyme mechanism — substrate, active site, product (B9) · the four levels of protein structure (B10) · peptide bond formation (B11) · plant vs animal cell (B6) · DNA double helix (B13b)
Unit 2	Competitive, non-competitive and uncompetitive inhibition (Q1) · alcoholic fermentation pathway (Q2) · chloroplast with thylakoid and Calvin cycle (Q4) · types of pesticides (Q6) · predators, parasites and pathogens in biocontrol (Q8) · how vaccines work (Q9)
Unit 3	Food chain and food web (Q11 short, Q3) · energy pyramid with the 10% law (Q4) · carbon cycle (Q5) · nitrogen cycle (Q6) · phosphorus and sulphur cycles (Q8) · oxygen cycle (Q9) · tolerance curve (Q10) · water cycle (Q11)
Unit 4	Acid rain formation cycle (Q2) · electrostatic precipitator (Q6) · flue gas desulfurisation (Q7) · loudness scale in dB (Q8) · cyclone separator and baghouse (Q9) · geosequestration of CO_2 (Q10)
Unit 5	BOD exertion curve (A5) · eutrophication stages (A6) · the four treatment phases (B1a) · trickling filter cross-section (B1b) · activated sludge process (B1c) · oxidation pond (B2) · the four elements of ISWM (B3) · hazardous waste plan (B4) · e-waste recycling steps (B5) · EM field regions (B7)

Sources: CY202 syllabus (5 units, no Unit 6); Unit 1, 2, 3 and 5 question banks; Unit 4 Air and Noise Pollution question bank; the ten model question papers; and the meeting summary. Diagrams are reproduced from those banks at 190 dpi; several are third-party illustrations (Encyclopaedia Britannica, US EPA and similar) as they appear in the source files.